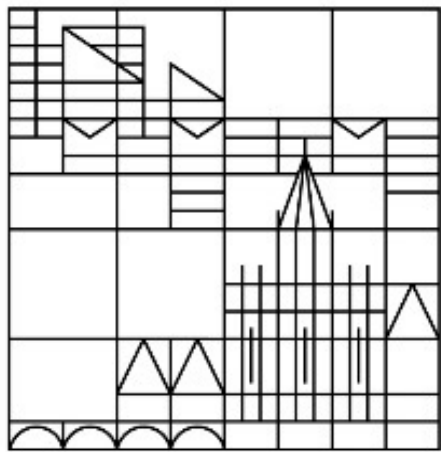




Show Me Your Best Side:

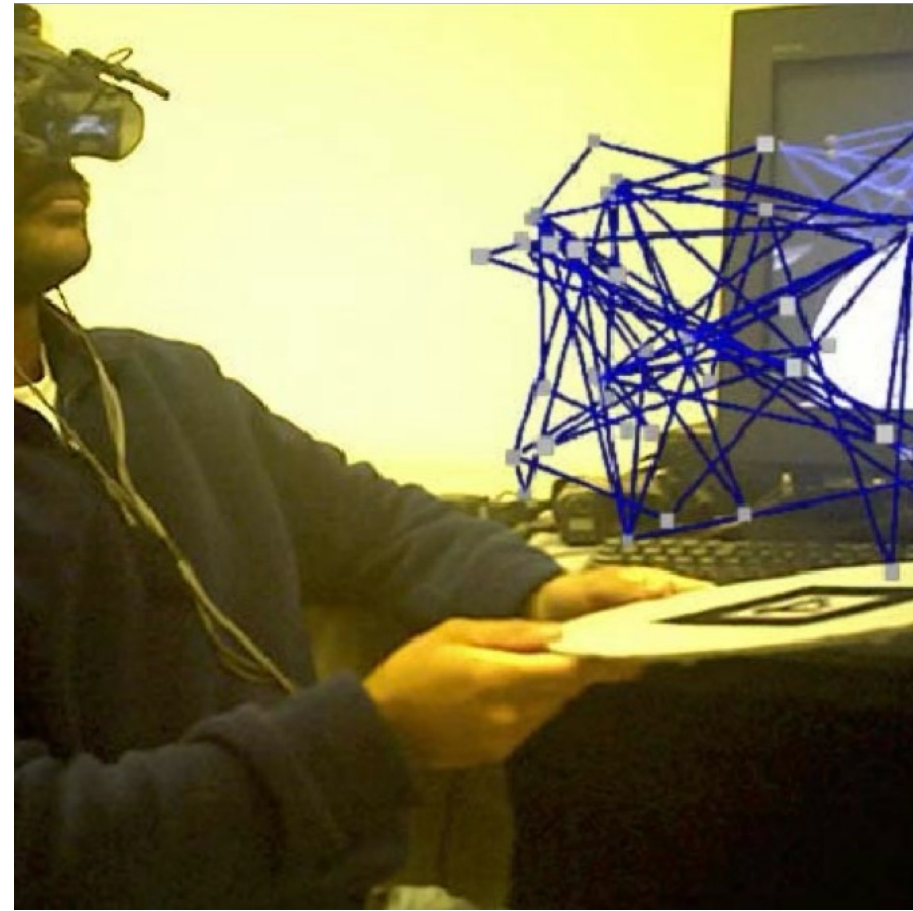
Characteristics of User-Preferred Perspectives for 3D Graph Drawings

Lucas Joos¹, Gavin J. Mooney², Maximilian T. Fischer¹, Daniel A. Keim¹,
Falk Schreiber^{1,2}, Helen C. Purchase², and Karsten Klein¹

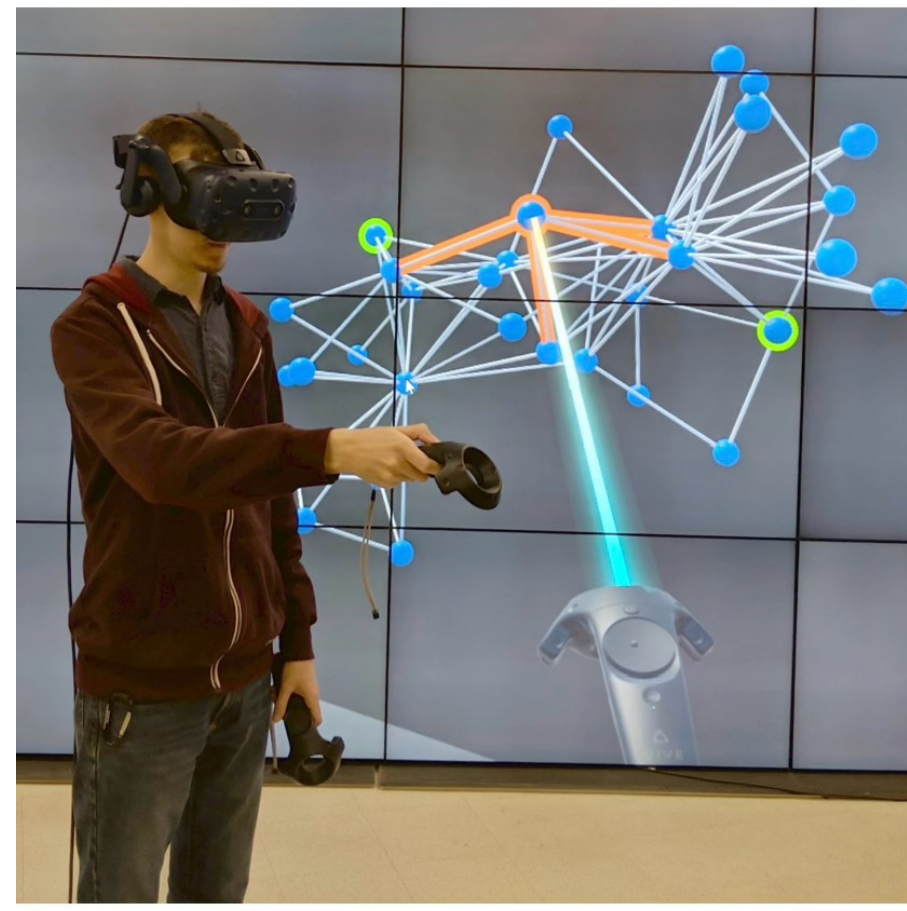
¹ University of Konstanz, Germany

² Monash University, Australia

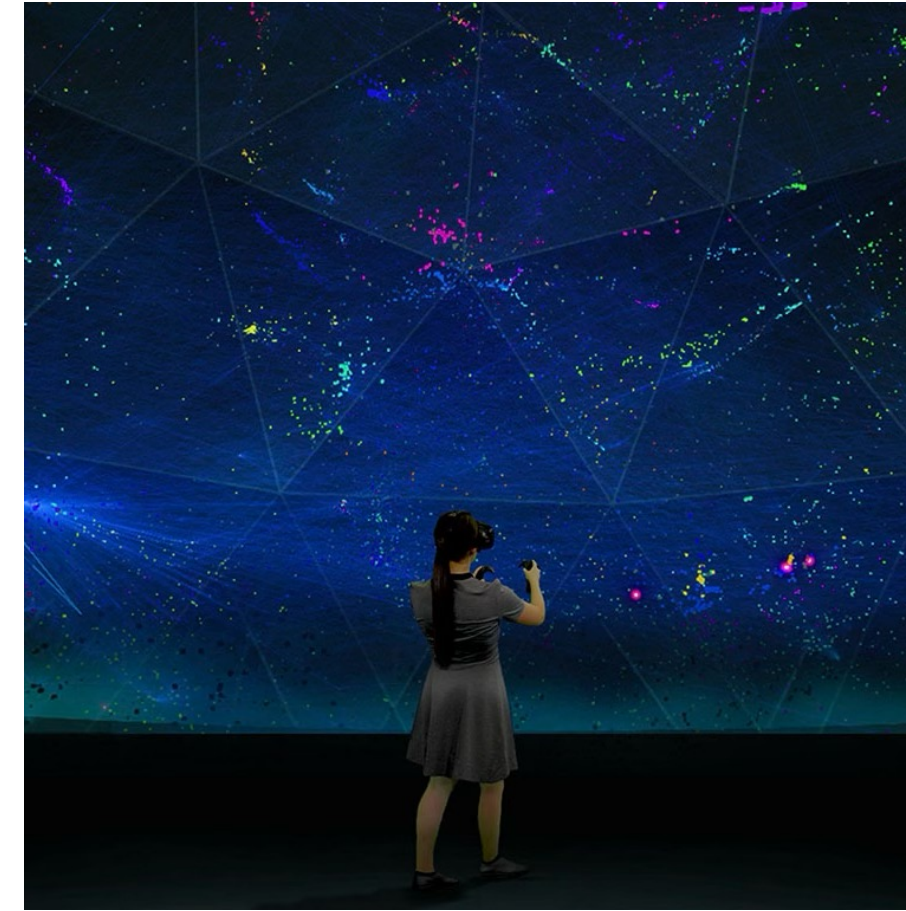
Motivation



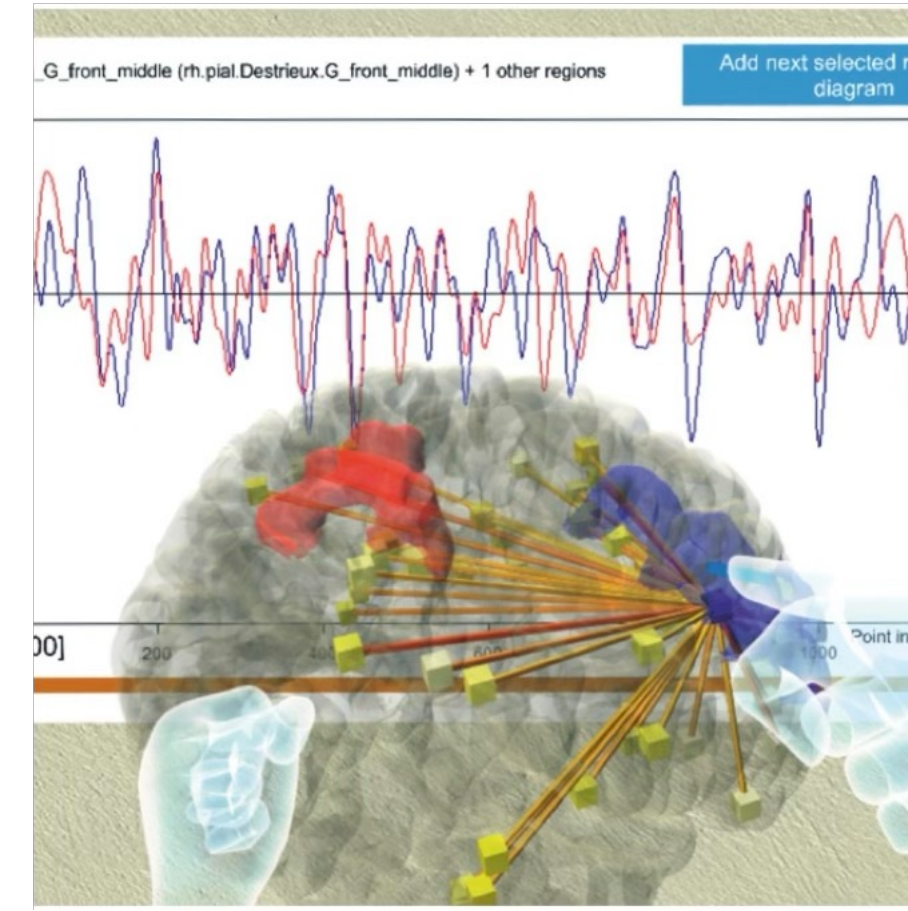
1



2



3



4



5

- ▶ Visual Network Analysis in Immersive Environments
 - ▶ Visual understanding capabilities and domain knowledge of experts
 - ▶ Realistic visualisation of graph data in the 3D context
 - ▶ Potentially more engaging, increases spatial understanding, task solving, natural interaction, collaborative work...

[1] Belcher et al. "Using augmented reality for visualizing complex graphs in three dimensions." Proc. ISMAR 2003. IEEE.

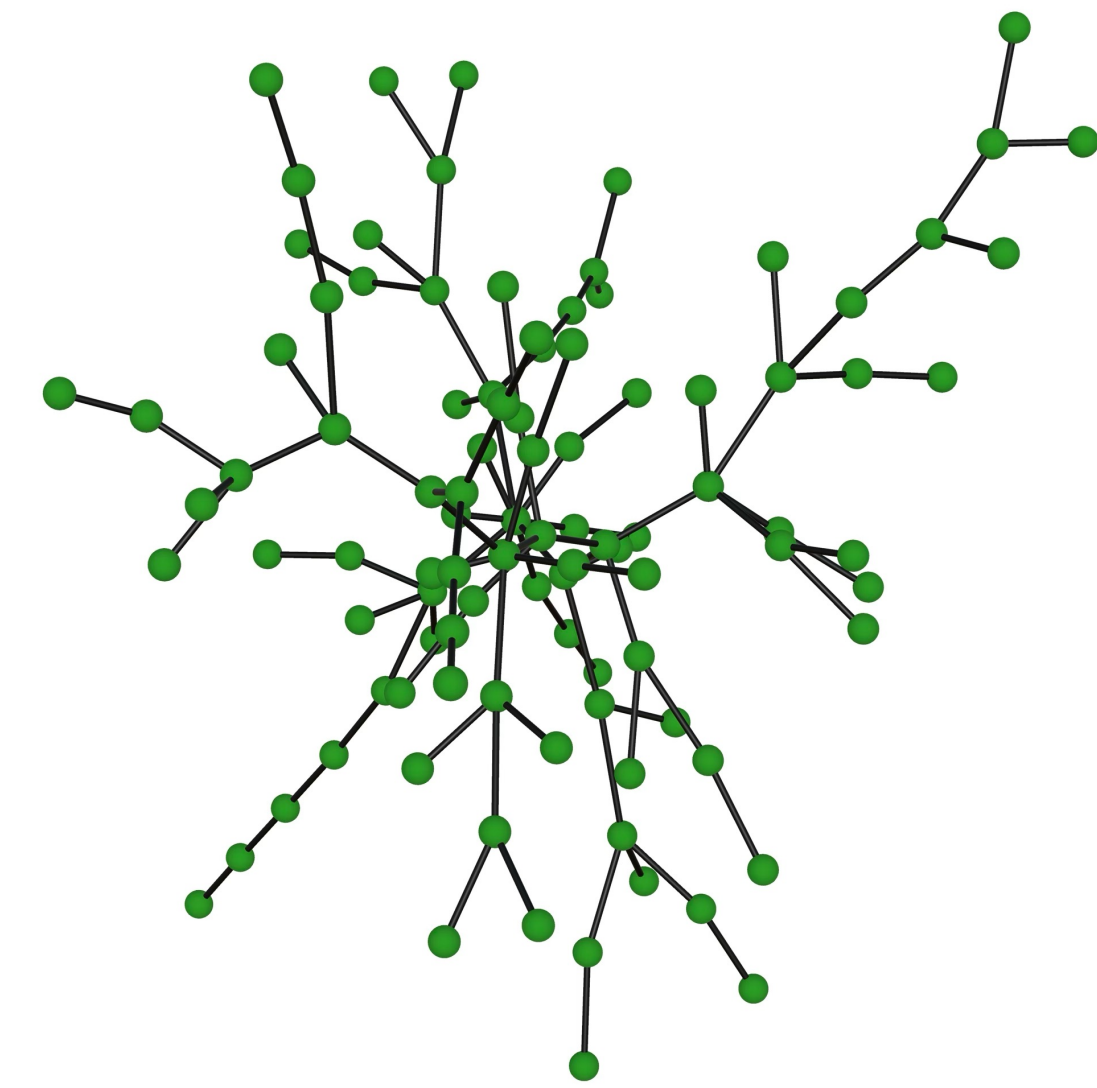
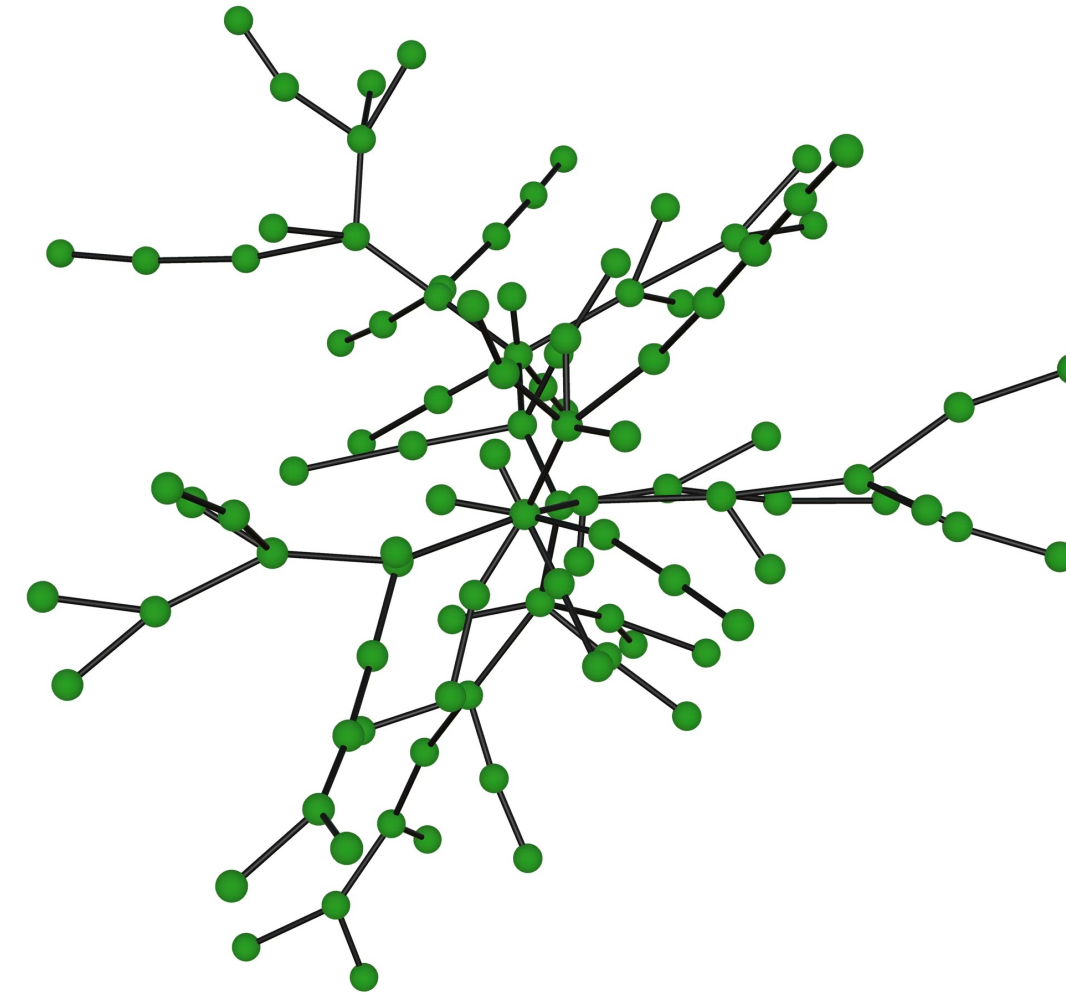
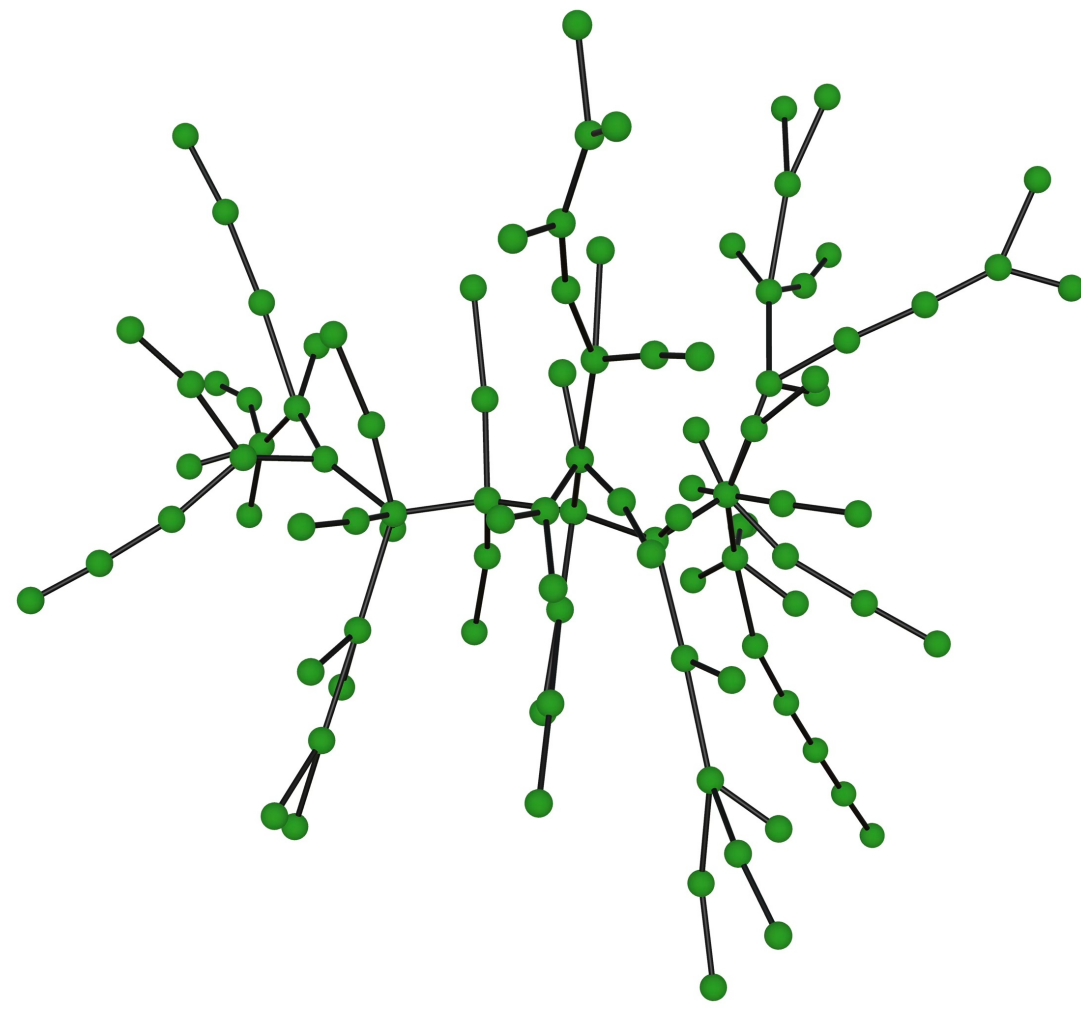
[2] Kotlarek et al. "A study of mental maps in immersive network visualization." Proc. PacificVis 2020. IEEE.

[3] Pirch et al. "The VRNetzer platform enables interactive network analysis in Virtual Reality." Nature Communications 12.1 (2021): 2432.

[4] Bauer et al. "A Multi-Layout Design for Immersive Visualization of Network Data." arXiv preprint (2021).

[5] Jaeger et al. "Challenges for brain data analysis in VR environments." Proc. PacificVis 2019. IEEE.

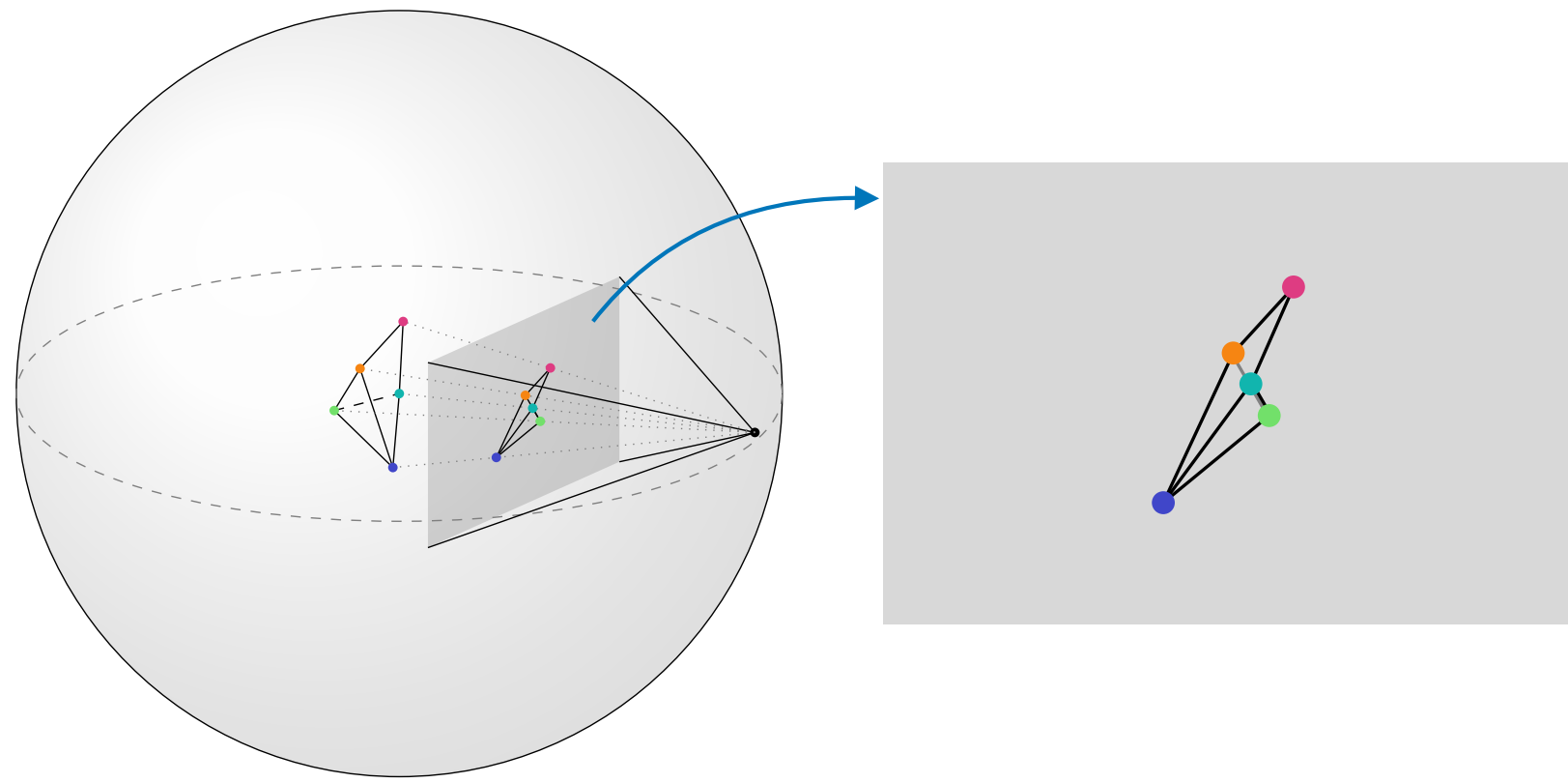
Motivation



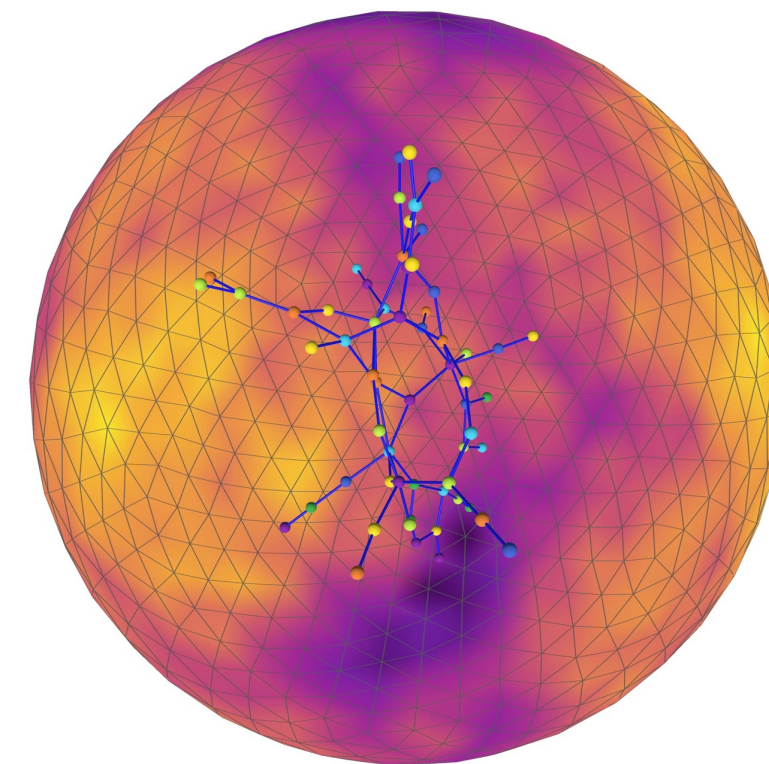
- ▶ Layout algorithms optimise for **entire** 3D graph
- ▶ Major challenge: **perspective**
- ▶ **Navigation** is required (manual)
- ▶ **Intuition**, which perspectives are better or worse

Motivation

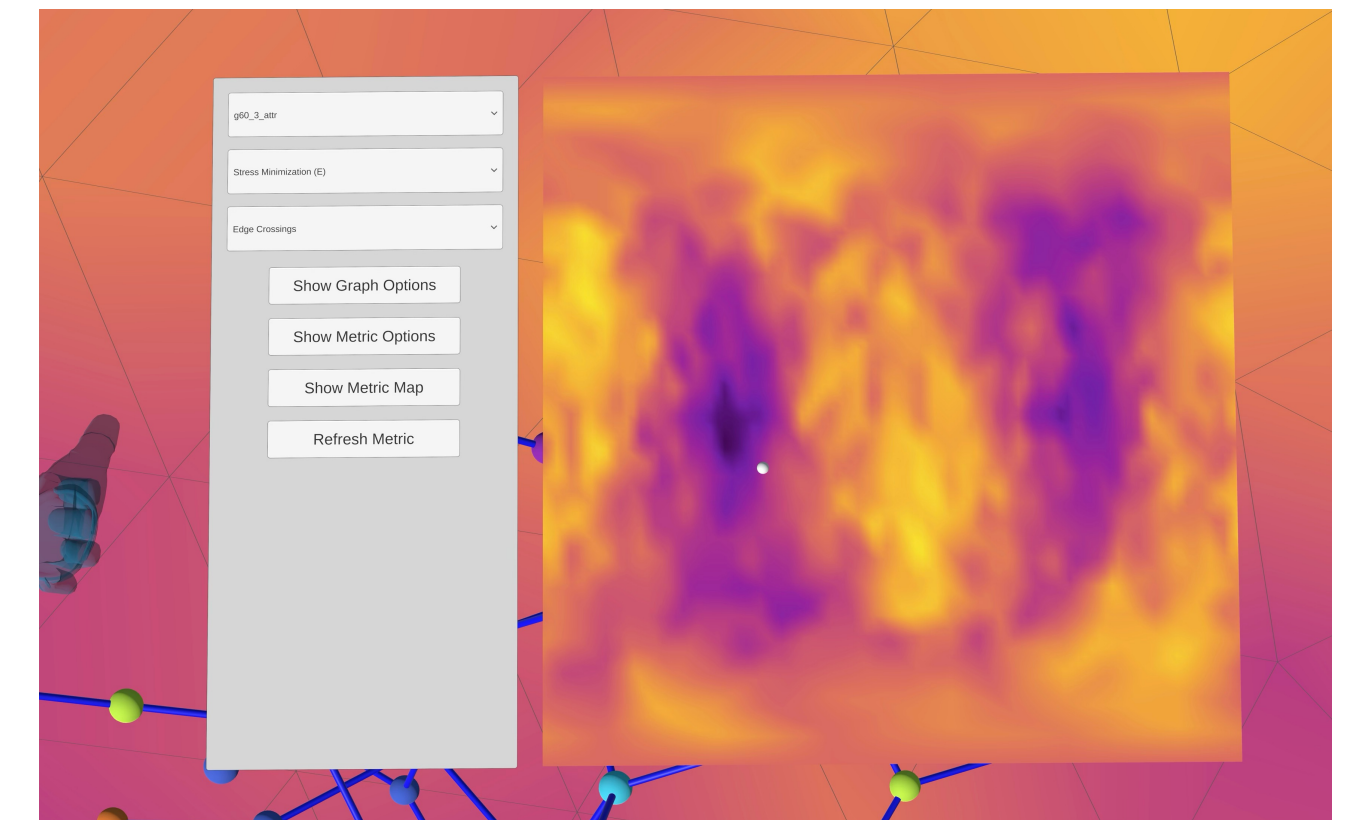
- ▶ **Previous Work:** Lucas Joos, Maximilian T. Fischer, Daniel A. Keim, and Johannes Fuchs. 2023. *Aesthetic-Driven Navigation for Node-Link Diagrams in VR*. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction (SUI '23)*. ACM, Article 25, 10 pages.



- 2D Projections for all Viewpoints
- Calculate Aesthetic Measures (Crossings, Crossing Angles, ...)



- Visual Representation of Measures



- Guided navigation

- ▶ Do 2D aesthetic measures also apply to graphs in **stereoscopic 3D**?
- ▶ How can we **quantify** how *good* a graph perspective in 3D is?
- ▶ Which **characteristics** are important for **users** to evaluate a perspective?

Research Question

- RQ: What are users' **preferences** regarding viewpoints in 3D graph drawings, and to what extent can individual **aesthetic measures** (e.g., edge crossings, stress) or **combinations** of these characterise preferred or unfavoured perspectives?
- Contributions:
 - VR **User Study** with 23 users and a broad set of graphs/layouts
 - Detailed **analysis** (quantitative and qualitative), empirically evaluating how well individual and combined **aesthetic measures** explain user choices
 - **Open-access** data for replicating and building up on the results

Methodology

- General Idea:
 - Data Collection: Show a broad set of graphs to users in an *immersive 3D* setup, ask them to find *best* and *worst* perspective. Also ask for *strategies*.
 - Evaluation: Check to what extent *aesthetic measures* (individual and combined) match user choices

Methodology: Data

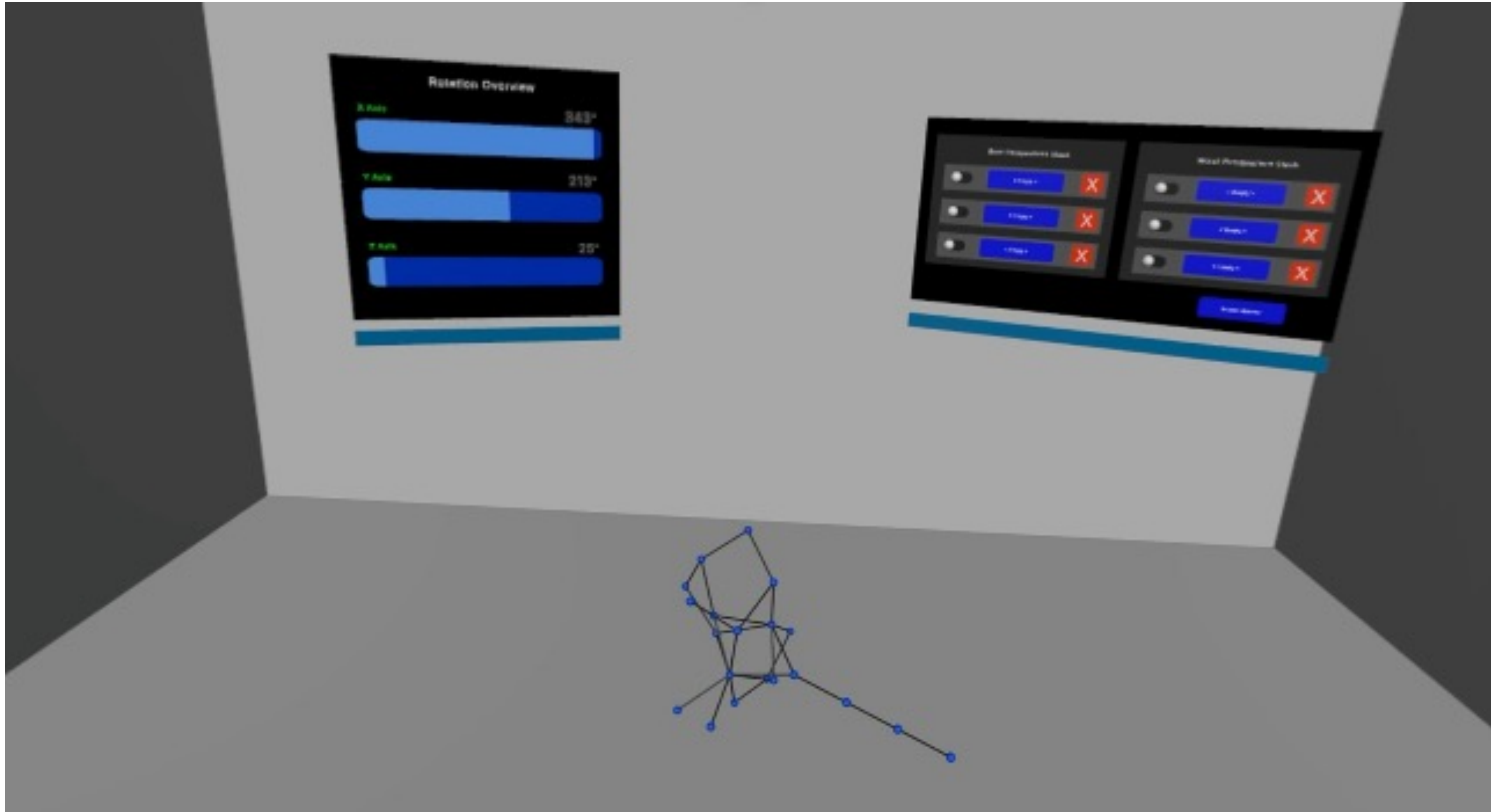
- Data (according to previous studies [6])
 - **36 graphs**: 4 sizes x 3 layouts x 3 graphs
 - Four **sizes** (numbers of nodes): S-S (20), S-M (50), S-L (100), and S-XL (200)
 - Three **layout/types**:
 - **Energy-Based (L-E)**:
Graphs from *ROME* dataset, *SuiteSparse Matrix Collection*, and *OGDF Random Tree Generator*, layouted with *OGDF Stress Minimisation* algorithm
 - **Layered (L-L)**:
Similar to [7], generating 2D graph layers (*OGDF Simple Connected Graph Generator*), and connecting layers randomly (varying number of edges)
 - **Semantic (L-S)**:
Brain activity networks (fMRI) with node positions corresponding to brain region centroids

[6] Joos et al. "Visual Network Analysis in Immersive Environments: A Survey." arXiv preprint arXiv:2501.08500 (2025).

[7] Feyer et al. "2D, 2.5 D, or 3D? an exploratory study on multilayer network visualisations in virtual reality." IEEE Transactions on Visualization and Computer Graphics 30.1 (2023): 469-479.

Methodology: Application

- Apple Vision Pro Application (Stereoscopic 3D)



Methodology: Procedure

- 23 users, randomized graph order
- Gaze+Pinch interaction
- Randomized initial perspective
- Ensured that users saw broad range of perspectives
- Quantitative results and questionnaire

Methodology: Aesthetic Measures

- 2D Aesthetic Measures (GdMetriX [8])

- Edge Crossings (CR)
- Stress (ST)
- Crossing Angular Resolution (CAR)
- Bounding Box Area (AR)
- Aspect Ratio (ASP)
- Node Concentration (CON)
- Node Orthogonality (NO)
- Gabriel Ratio (GR)
- Angular Resolution (ANGR)
- Edge Orthogonality (EO)
- Edge Length Deviation (ELD)
- Edge Reflective Symmetry (ESR)
- Edge Rotational Symmetry (ESO)
- Edge Translational Symmetry (EST)

Methodology: Aesthetic Measures

- 3D Aesthetic Measures

- Node-Node Overlap Count ([NNO](#))
- Edge-Node Overlap Count ([ENO](#))
- Node-Edge Overlap Count ([NEO](#))
- Node-Node Overlap Area ([NNOA](#))
- Edge-Node Overlap Area ([ENOA](#))
- Node-Edge Overlap Area ([NEOA](#))

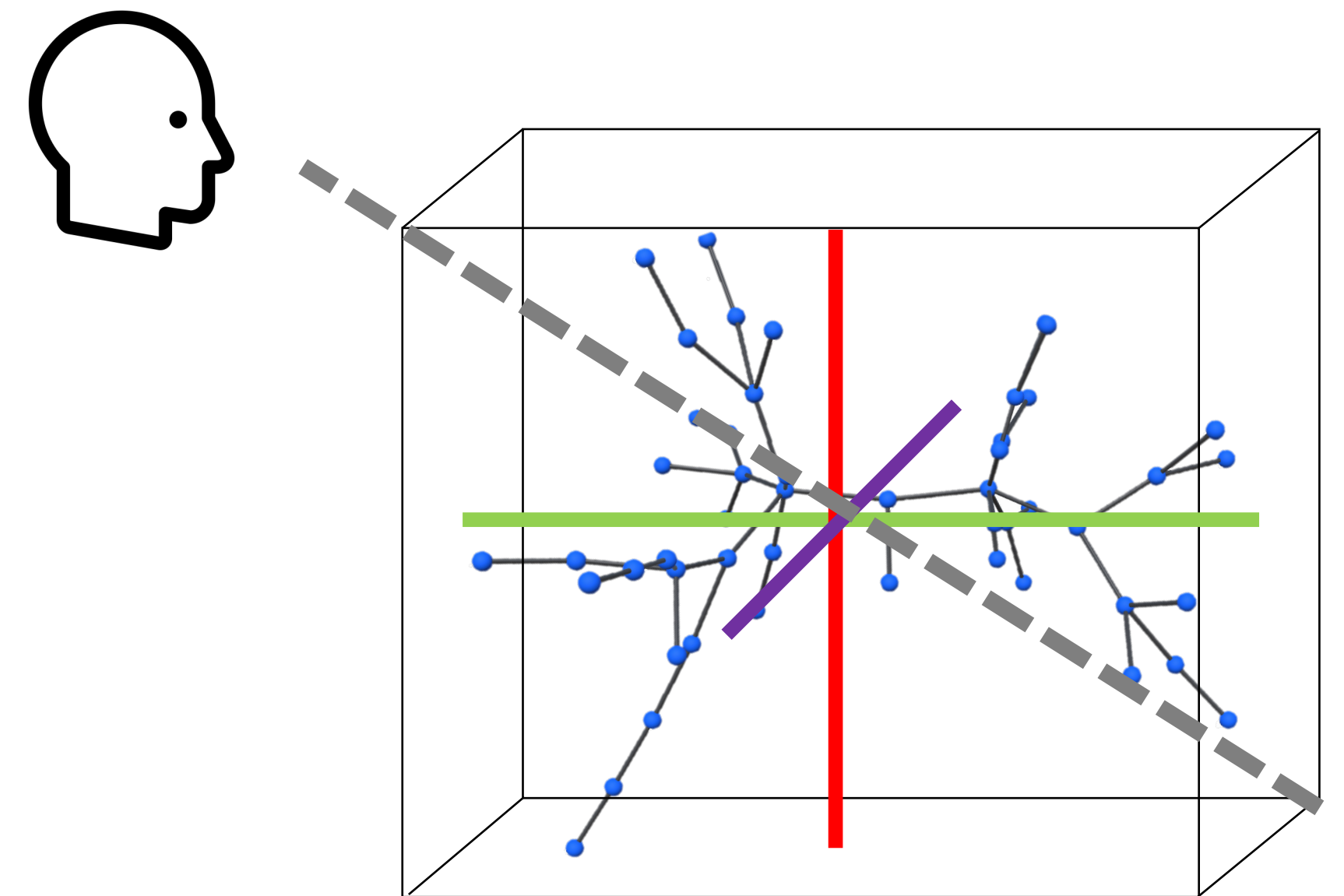
Methodology: Aesthetic Measures

- A new measure: **Isometric Viewpoint Deviation** (ISO)
- Main Idea: If the graph has meaningful **axes**, best way is to have an **isometric** view on them
- **PCA** on nodes for finding main dimensions (Eigenvectors) and the variance distribution (Eigenvalues)
- Evaluate how much a perspective **deviates** from the **isometric** view on these axes
- Not meaningful in all cases (e.g. spherical layouts)
- $ISO = 0$: graph has meaningful **main axes** and view is far from **isometric**
- $ISO = 1$: graph does not have meaningful **main axes** or otherwise, view is **isometric**

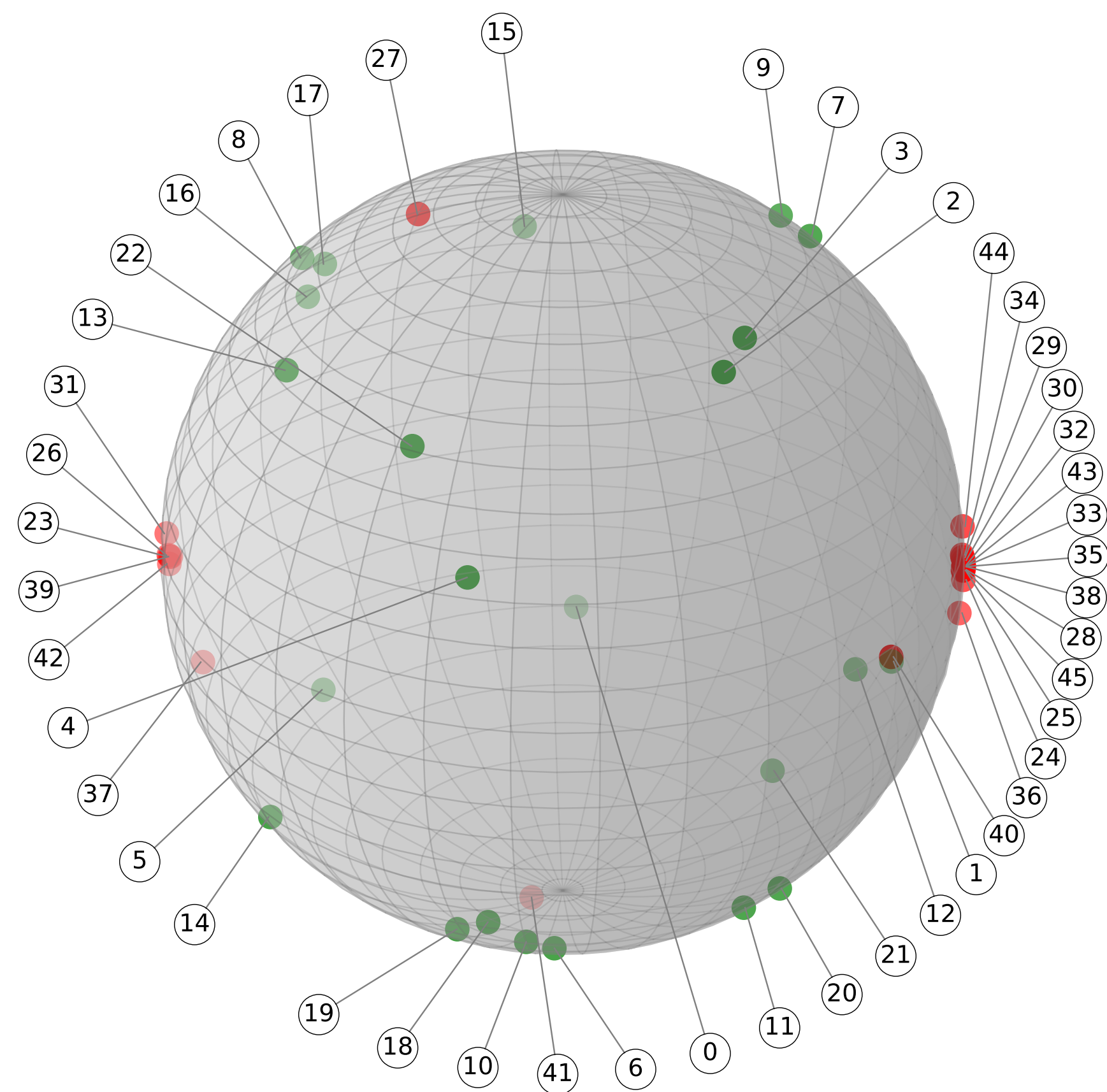
$$ISO = 1 - \alpha \cdot \frac{\sigma_w}{\sigma_{max}}$$

How much do the normalised Eigenvalues deviate?

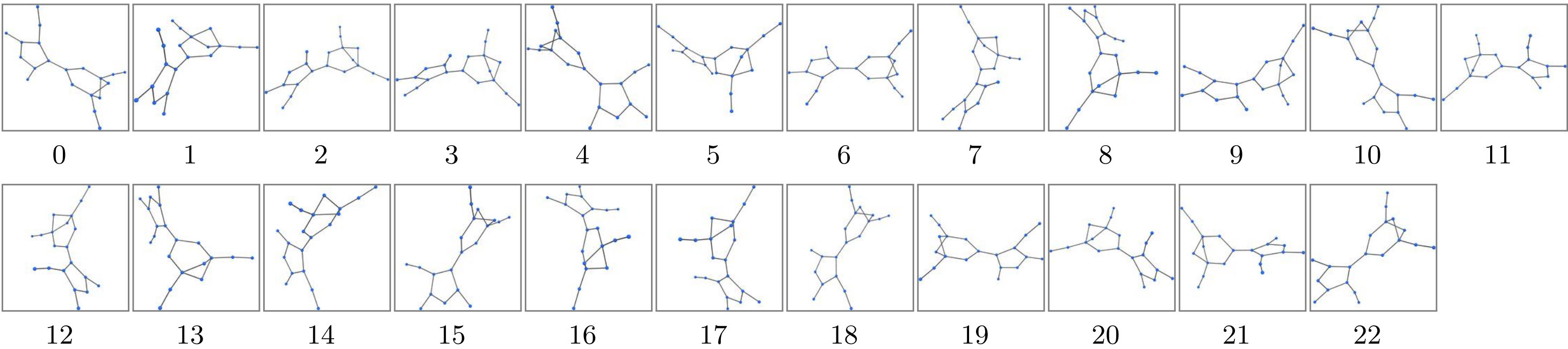
How imbalanced is the view w.r.t. the Eigenvectors?



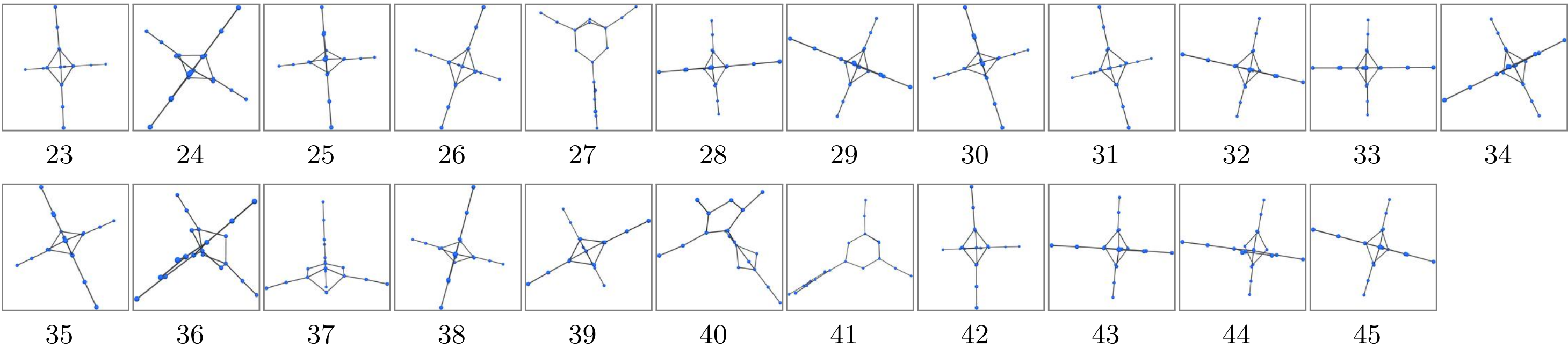
Results: Examples (1)



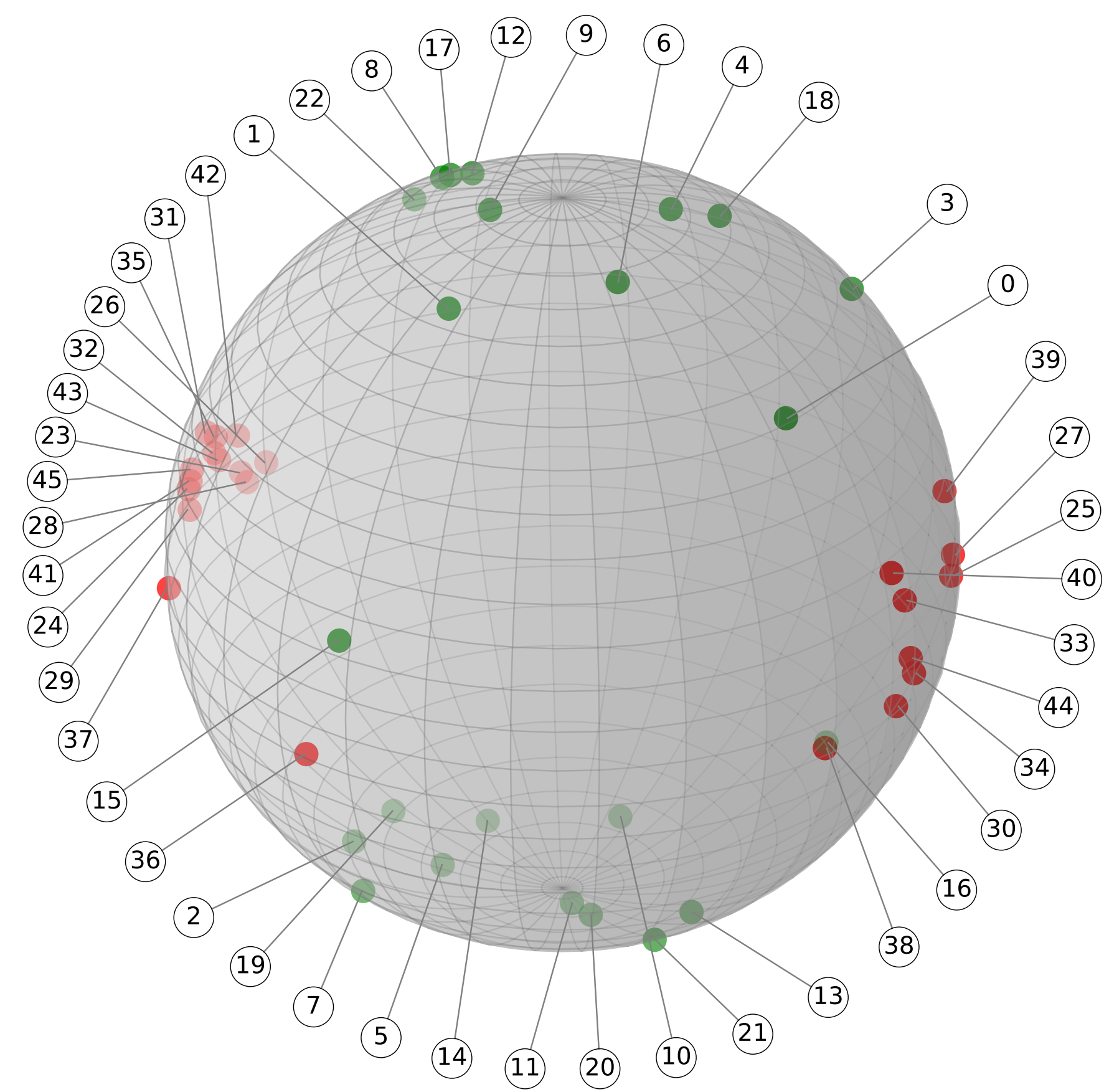
Best



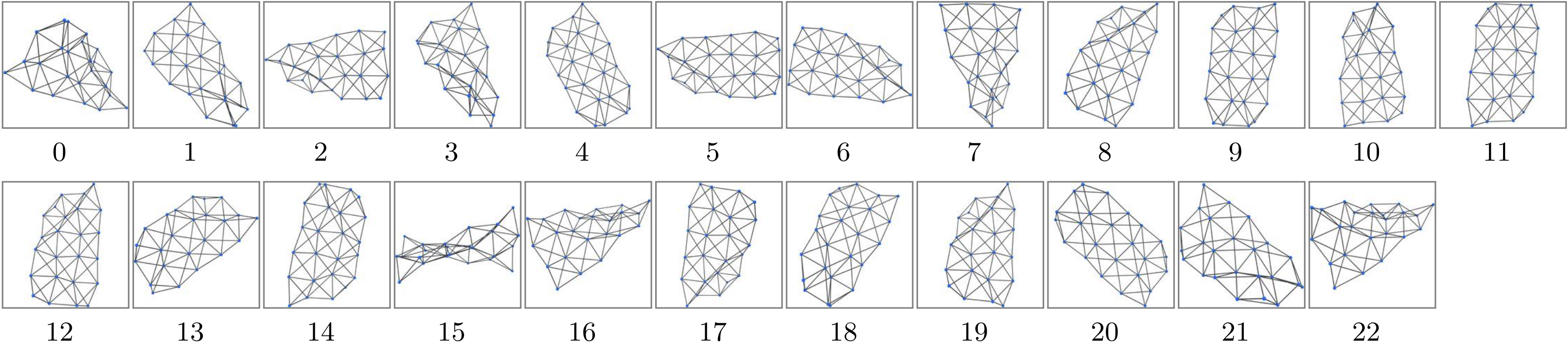
Worst



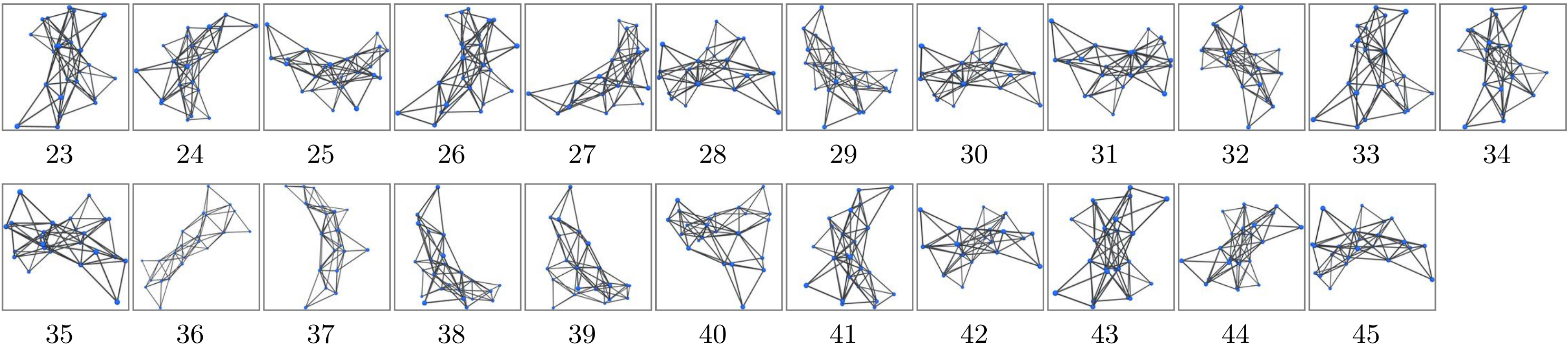
Results: Examples (2)



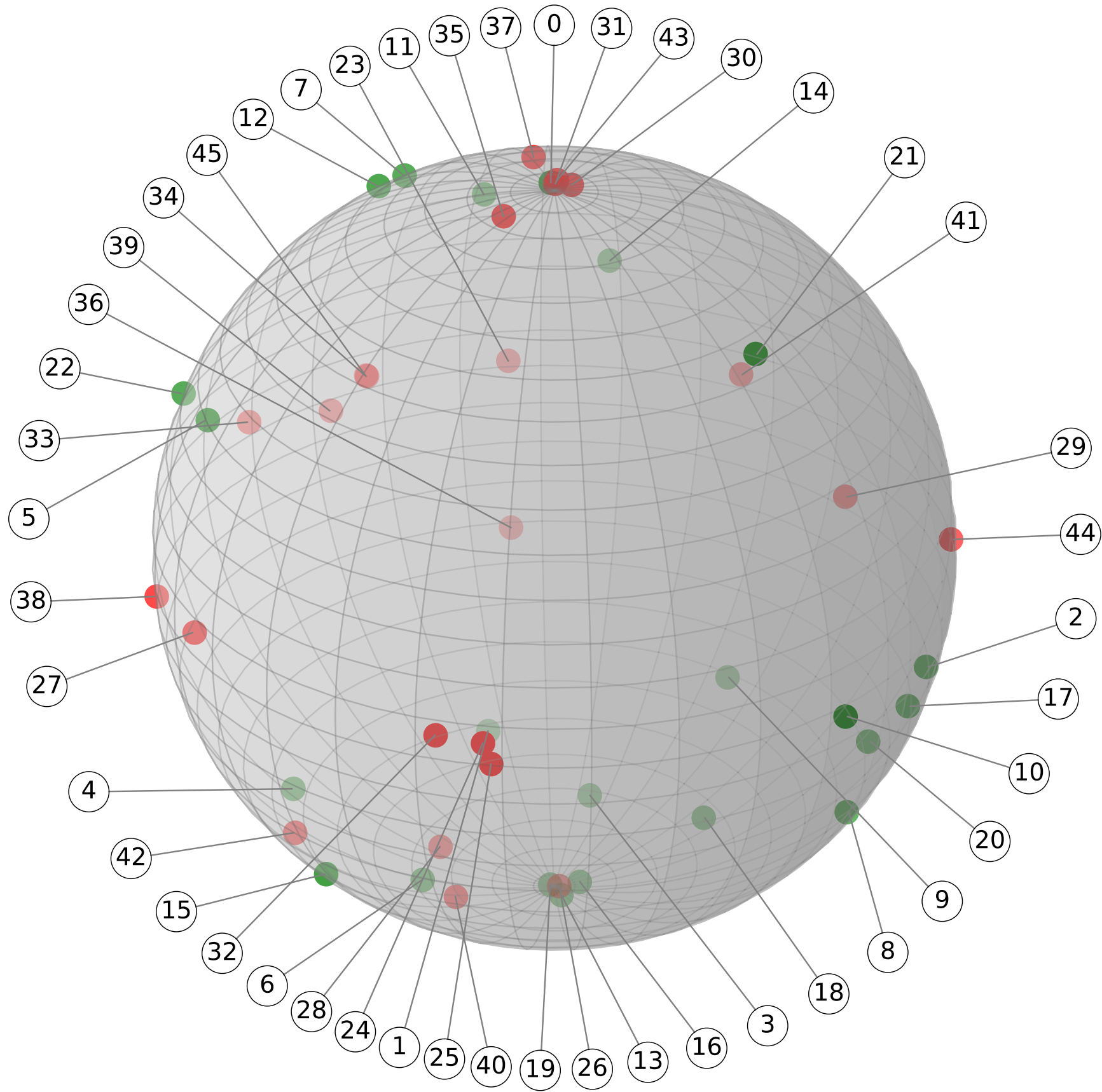
Best



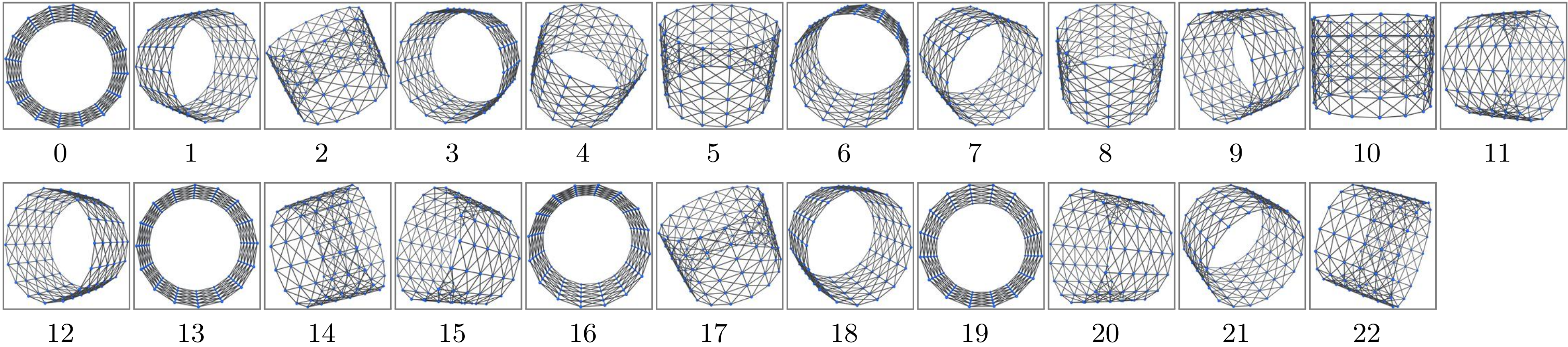
Worst



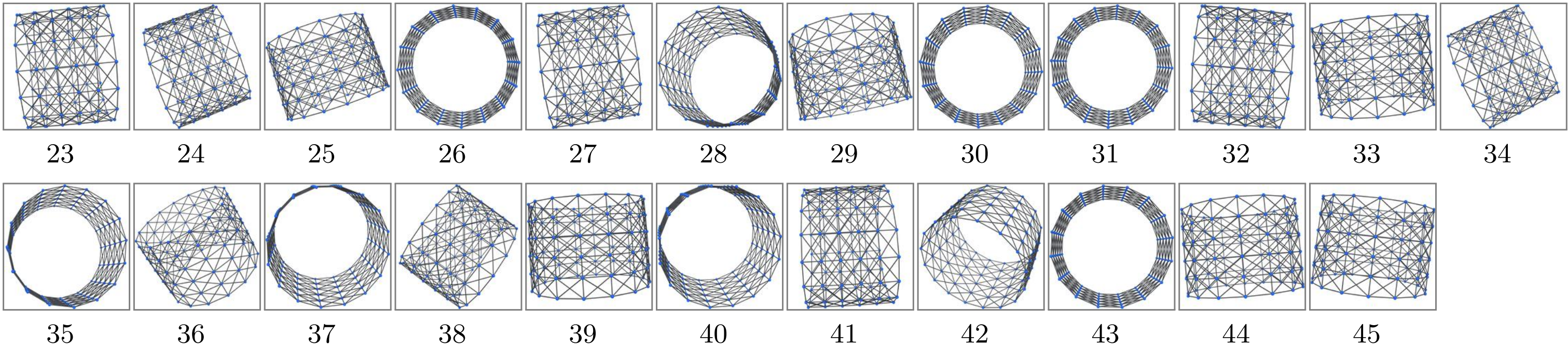
Results: Examples (3)



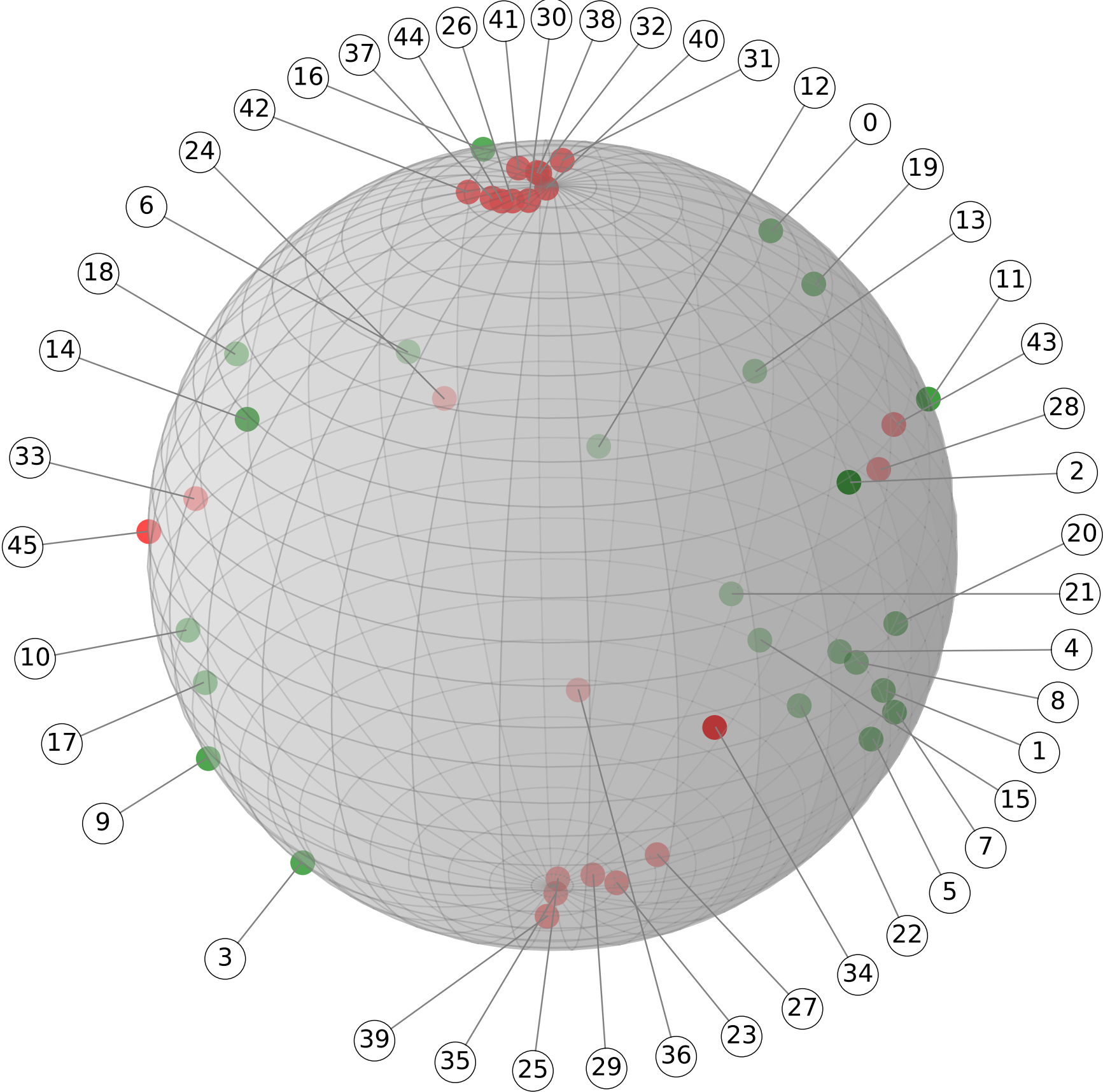
Best



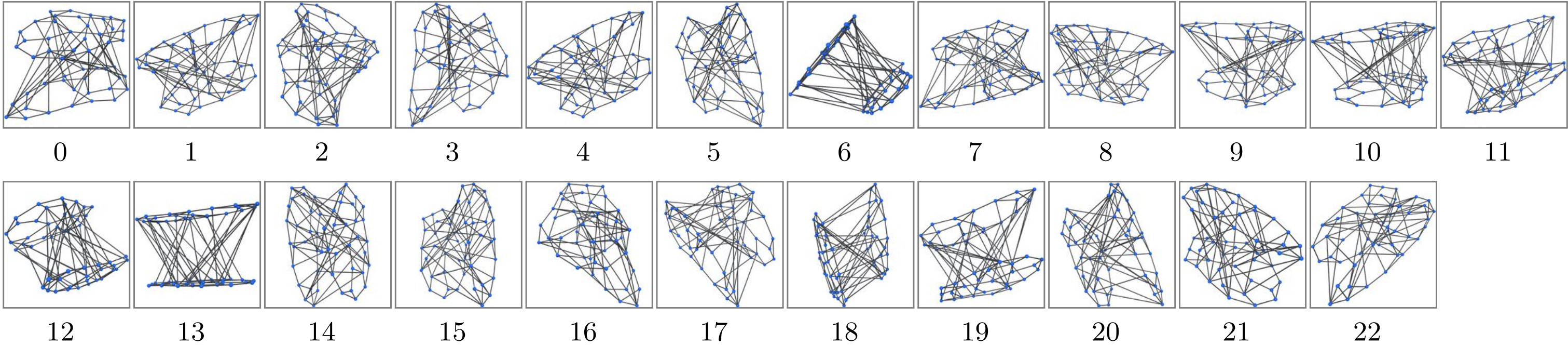
Worst



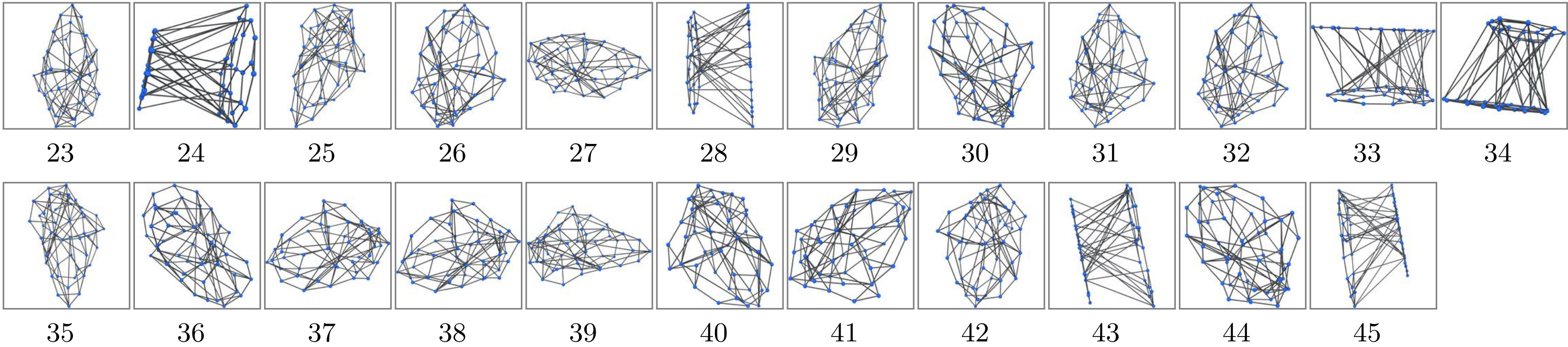
Results: Examples (4)



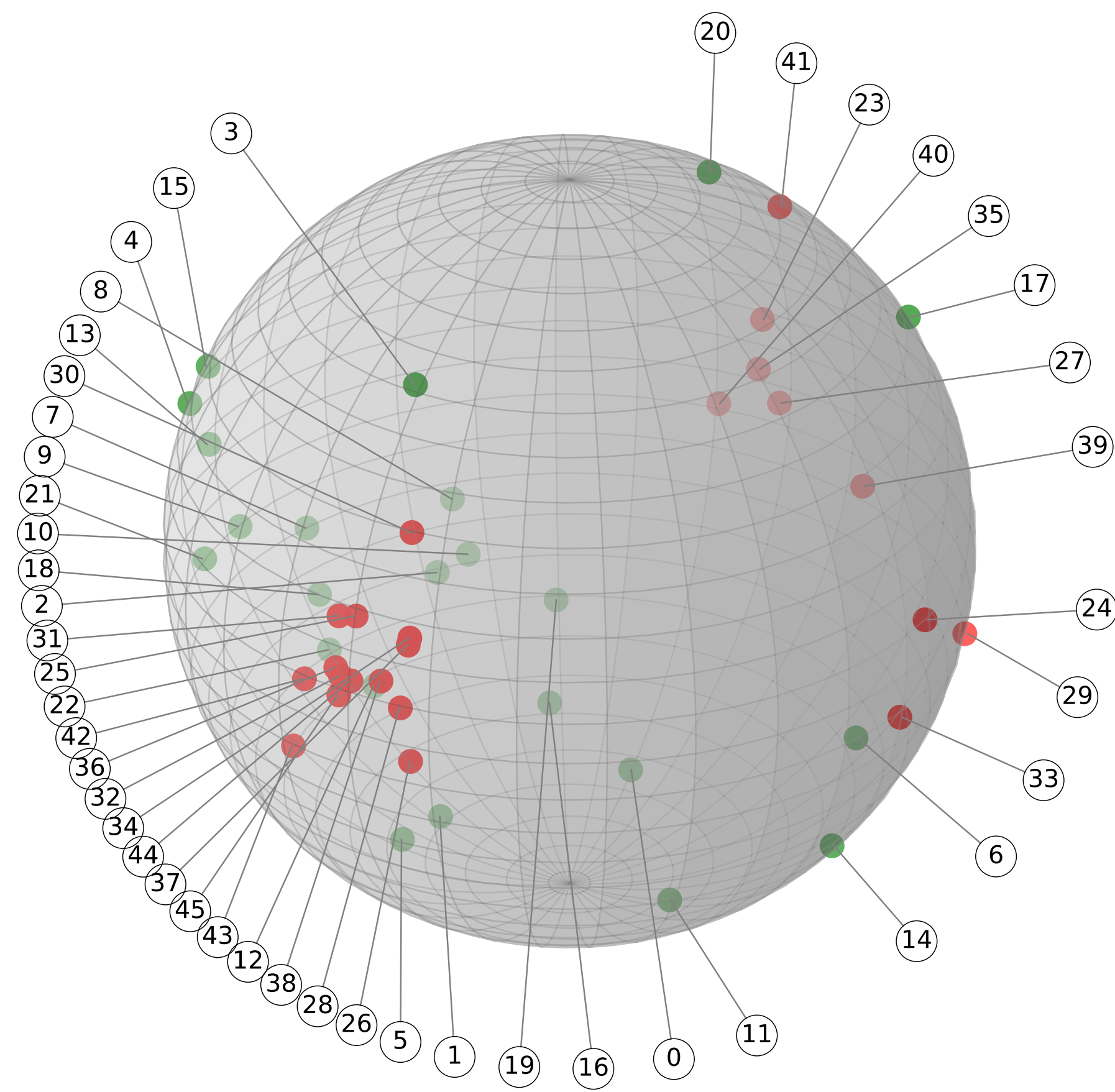
Best



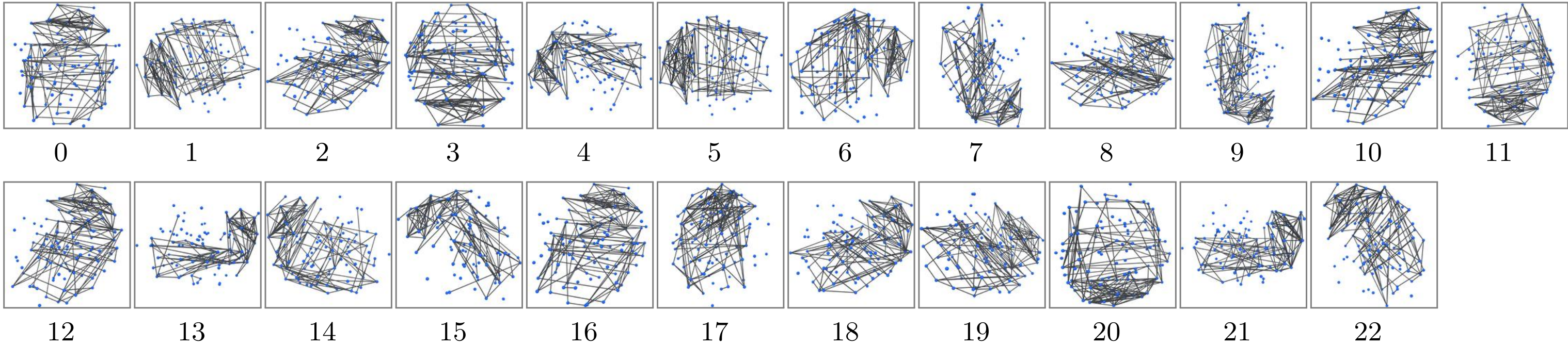
Worst



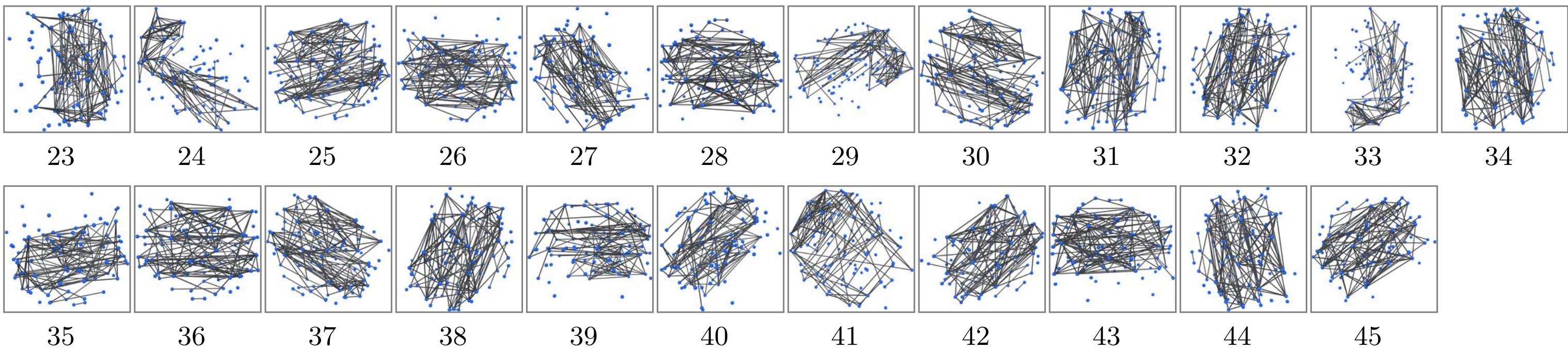
Results: Examples (5)



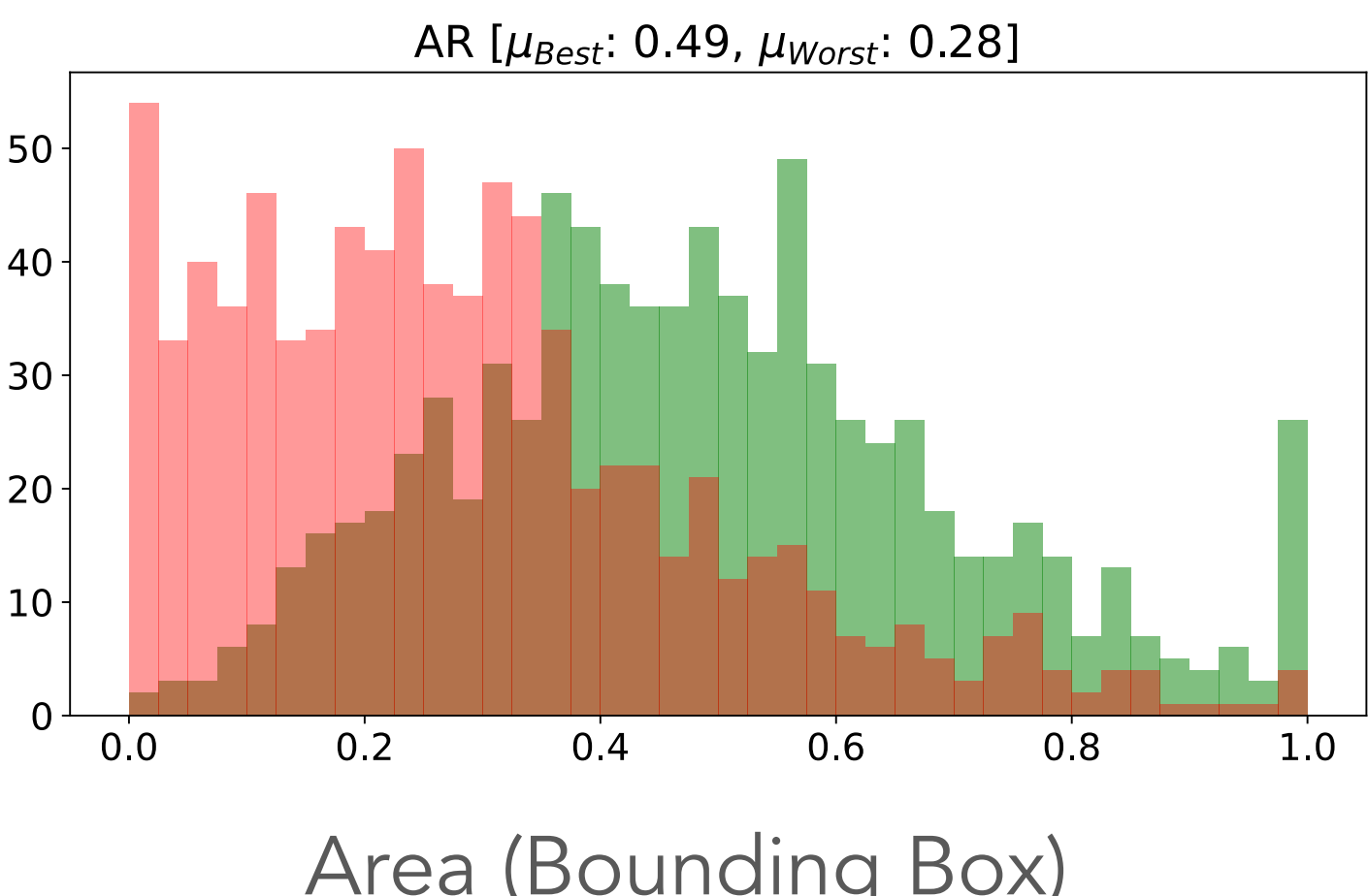
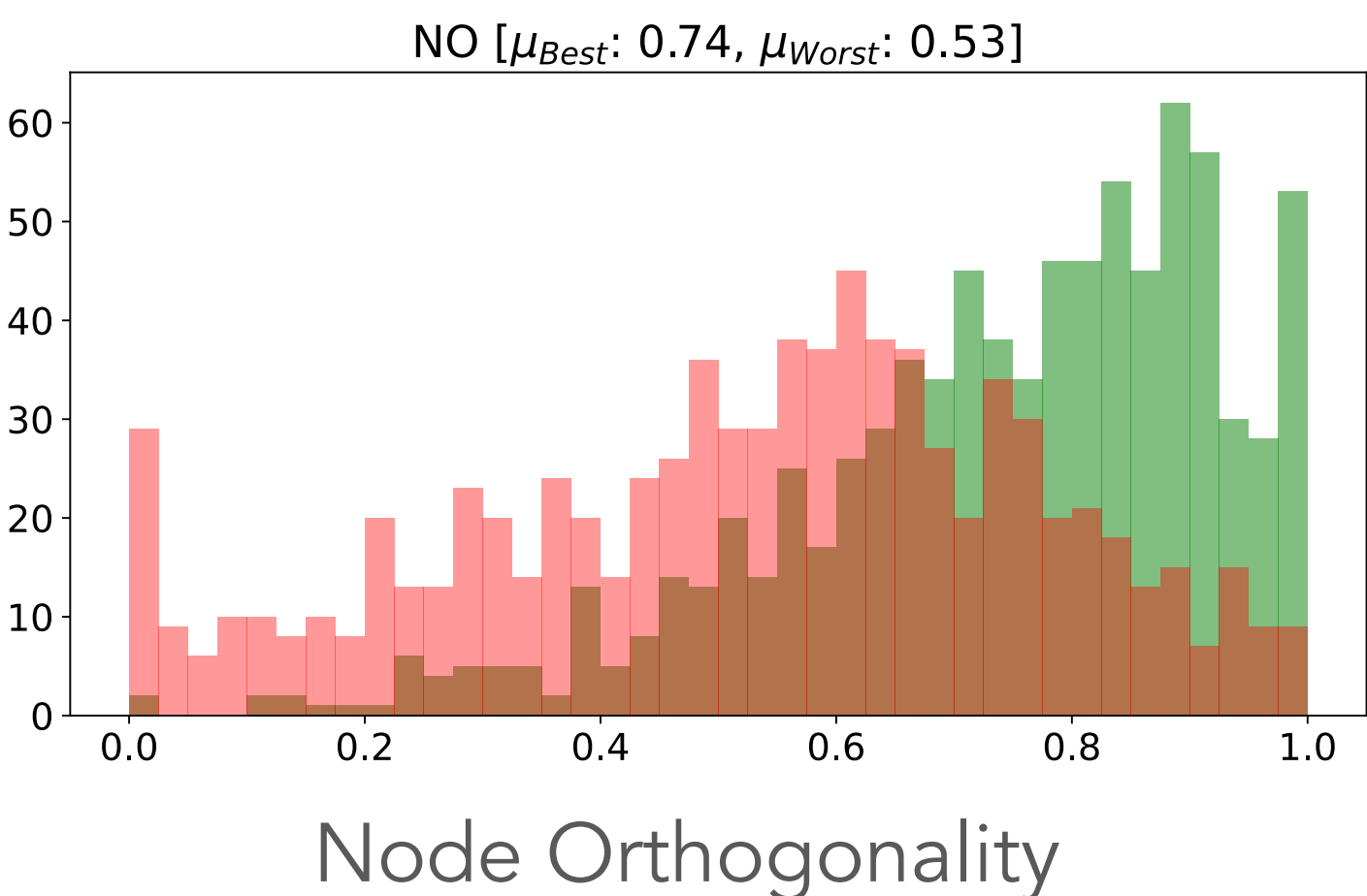
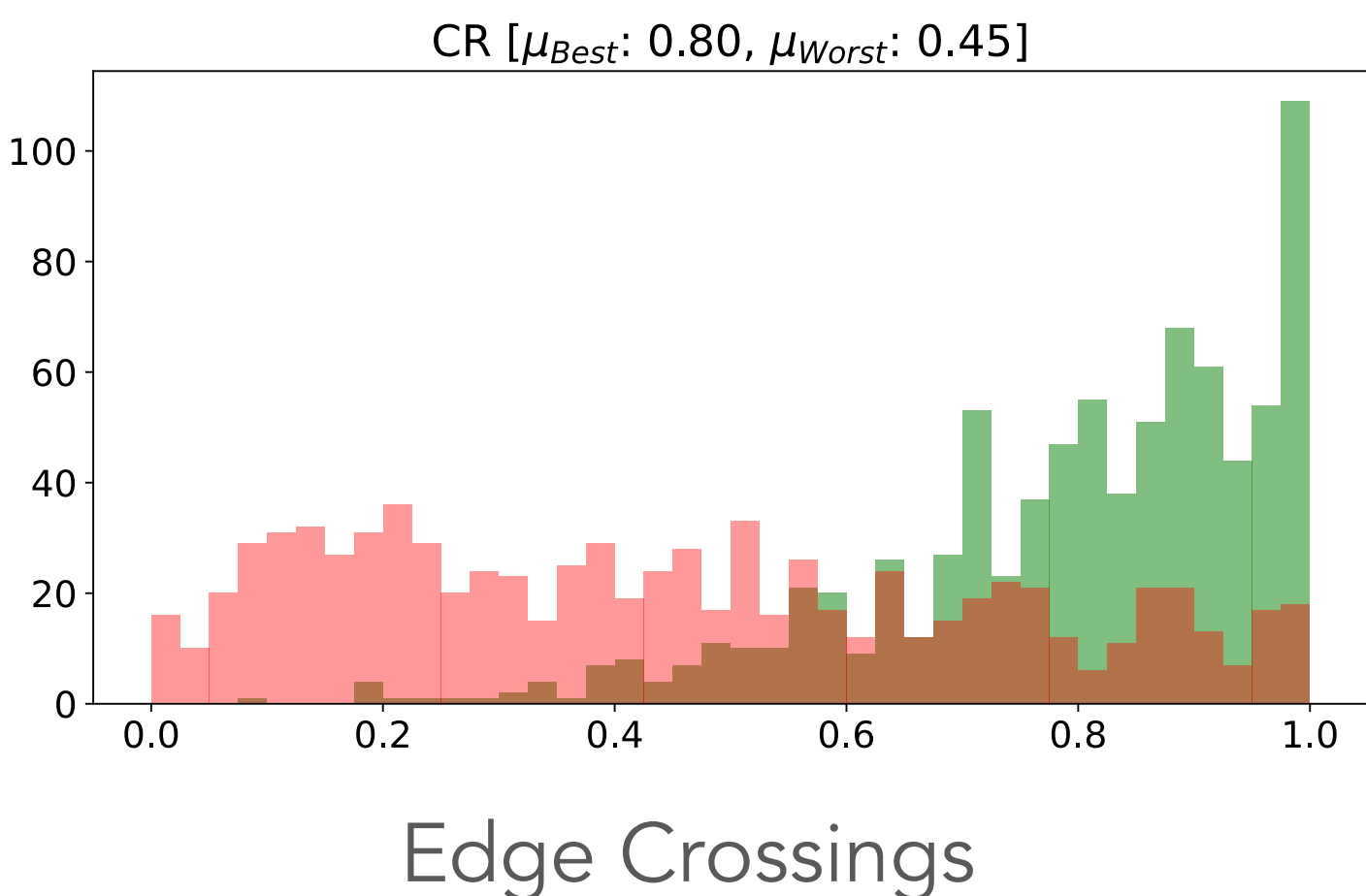
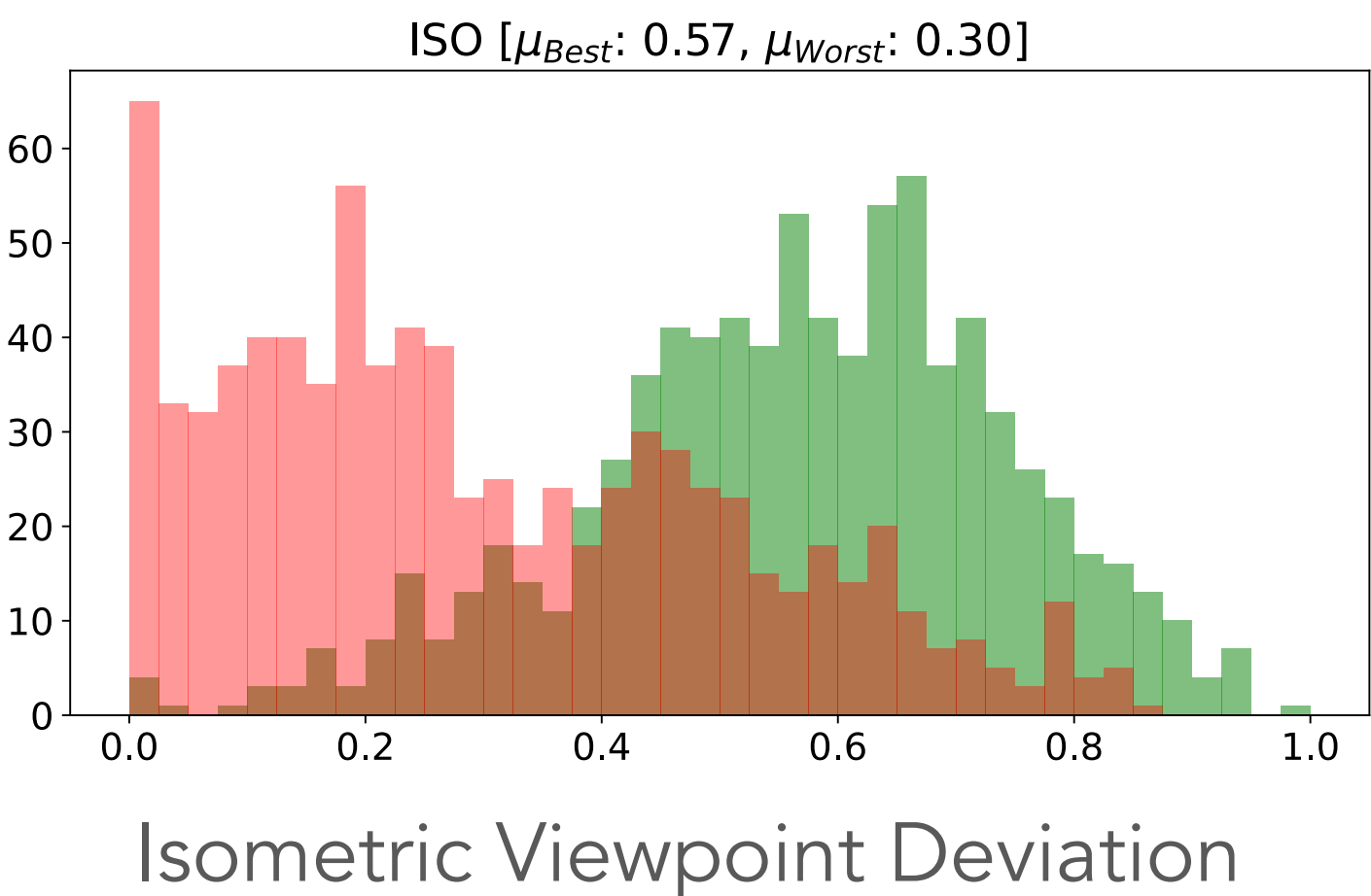
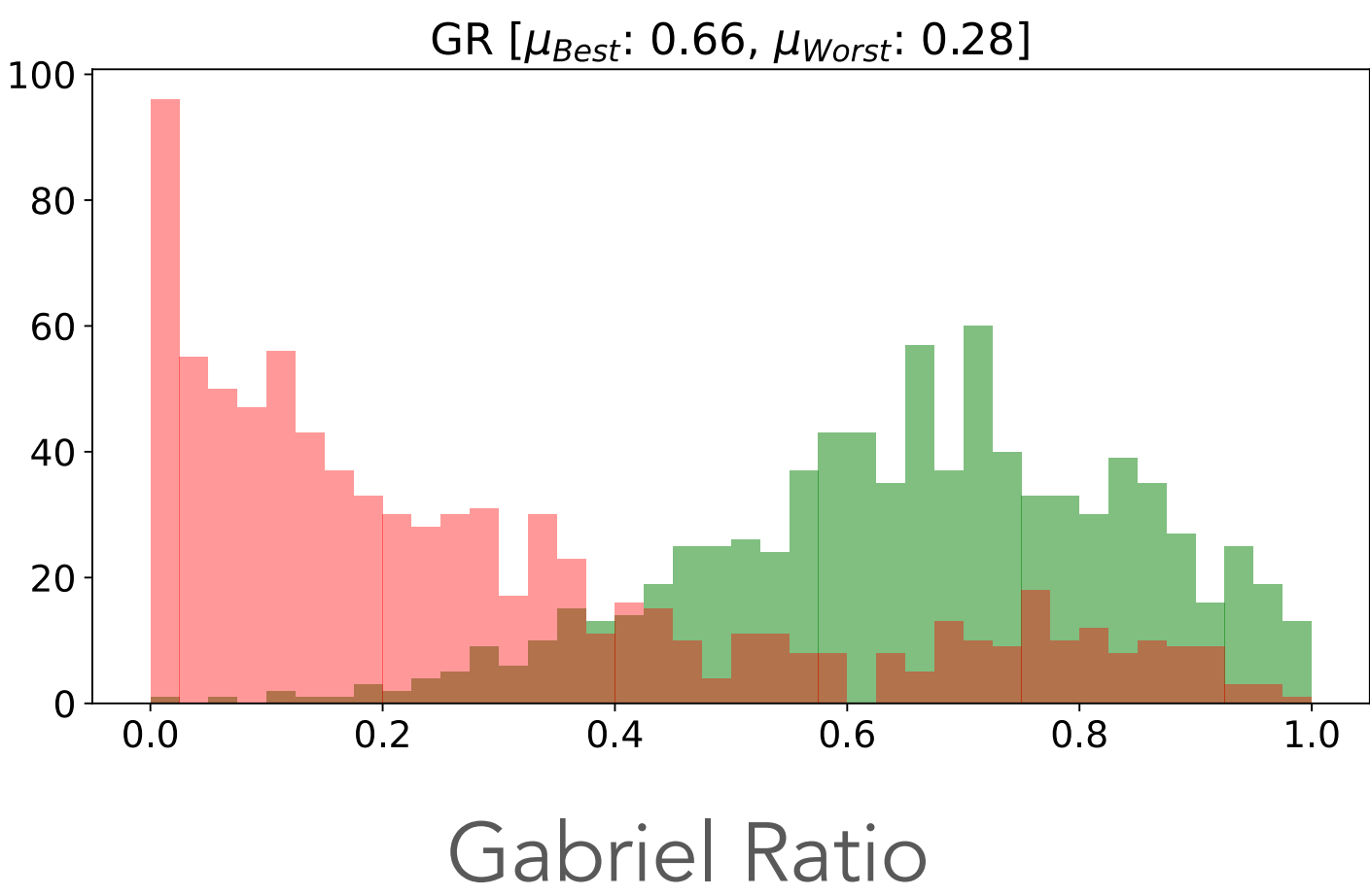
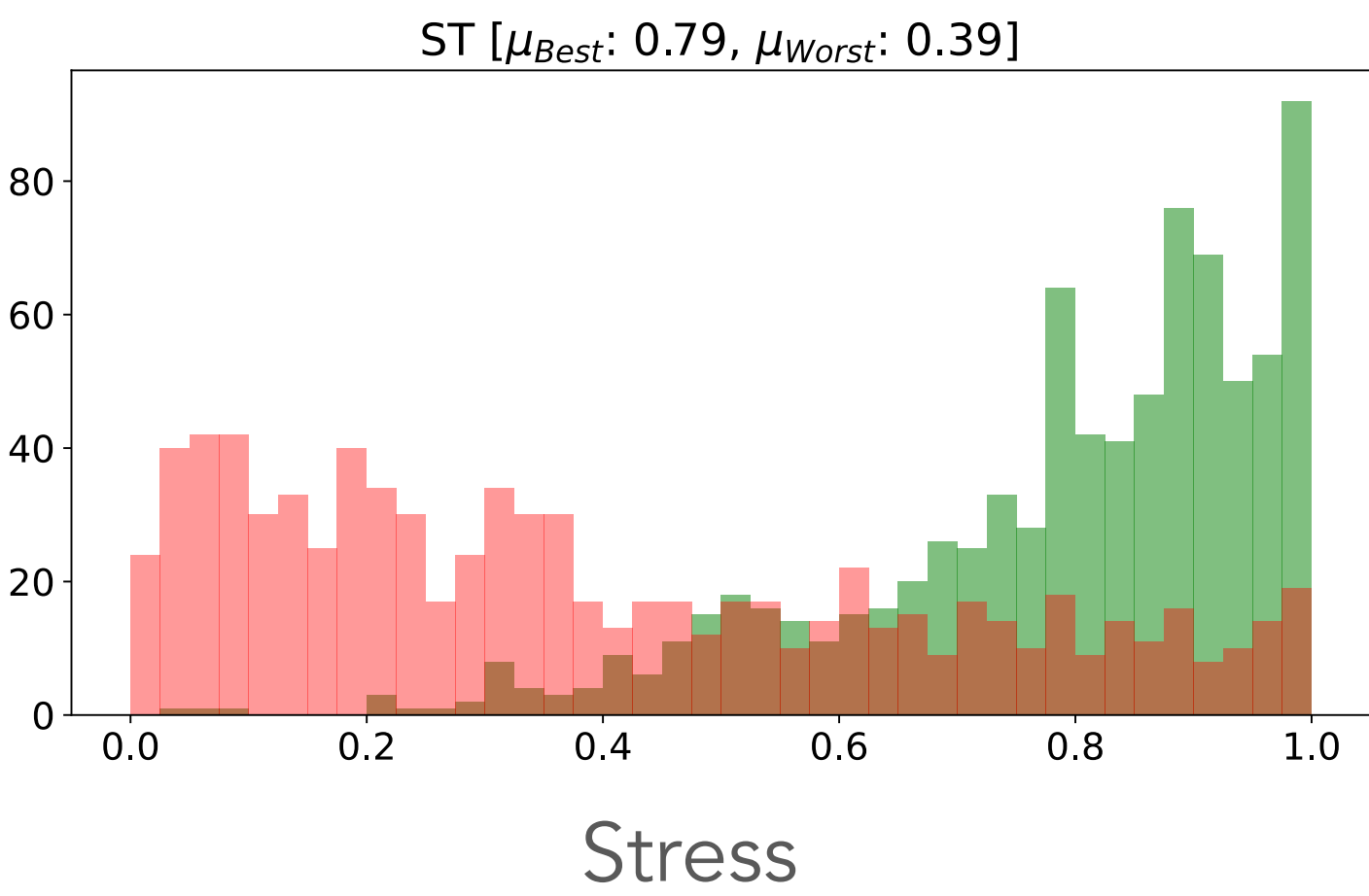
Best



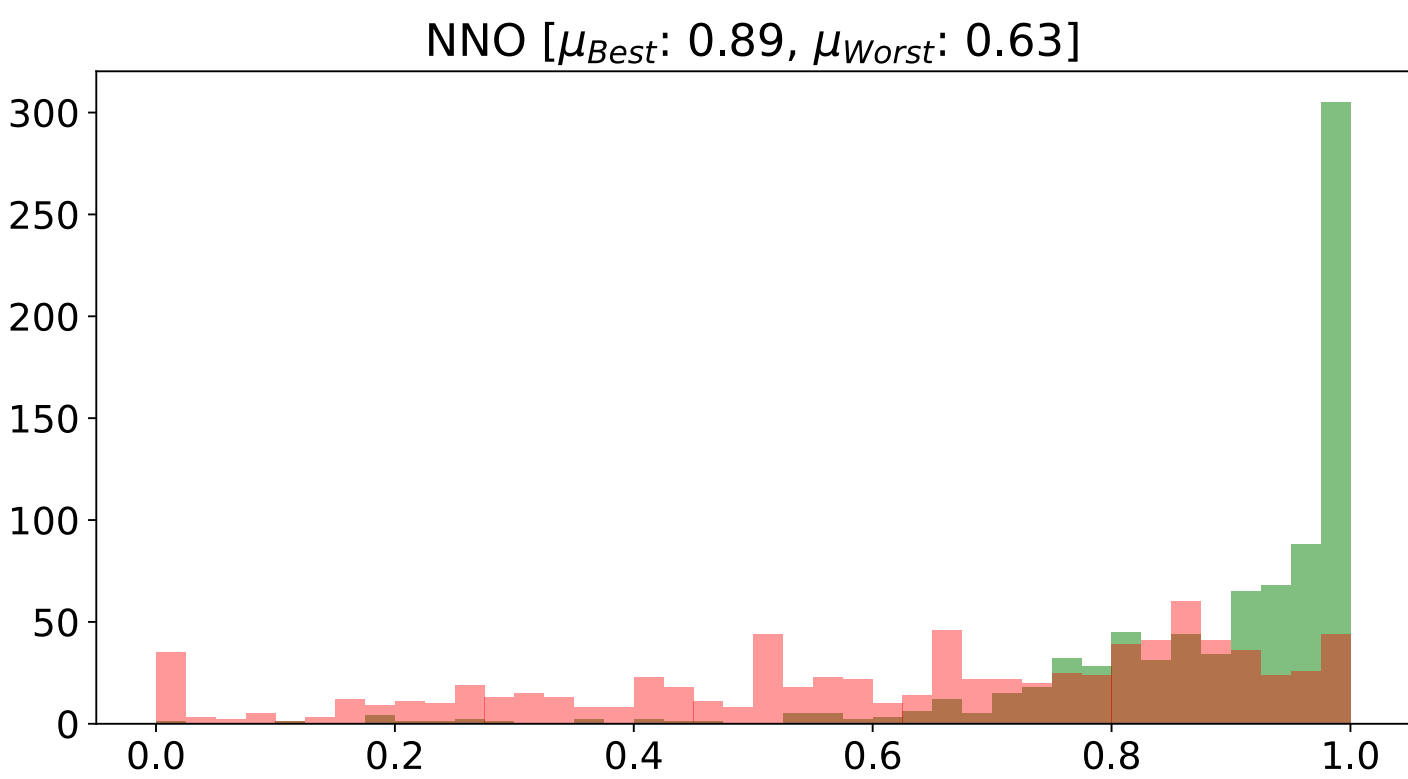
Worst



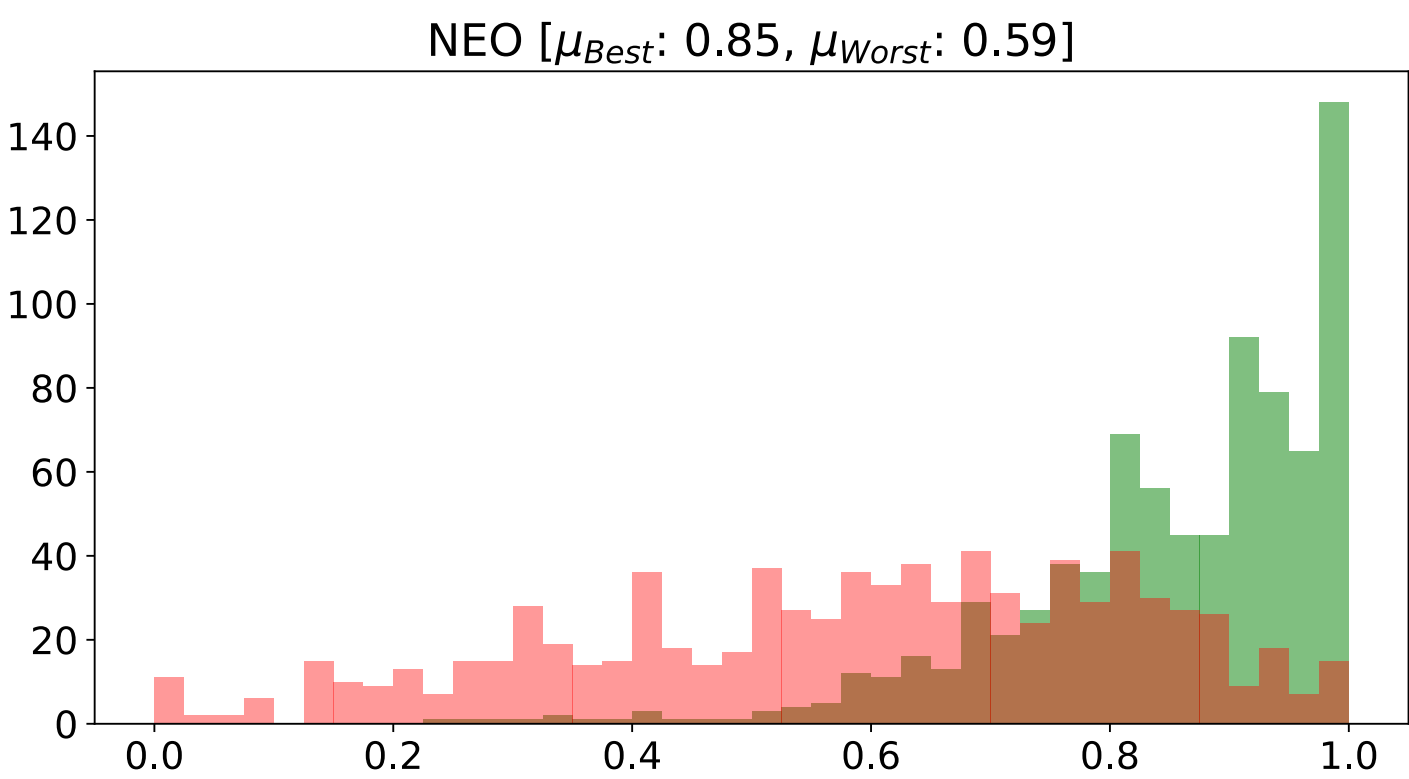
Results: Aesthetics



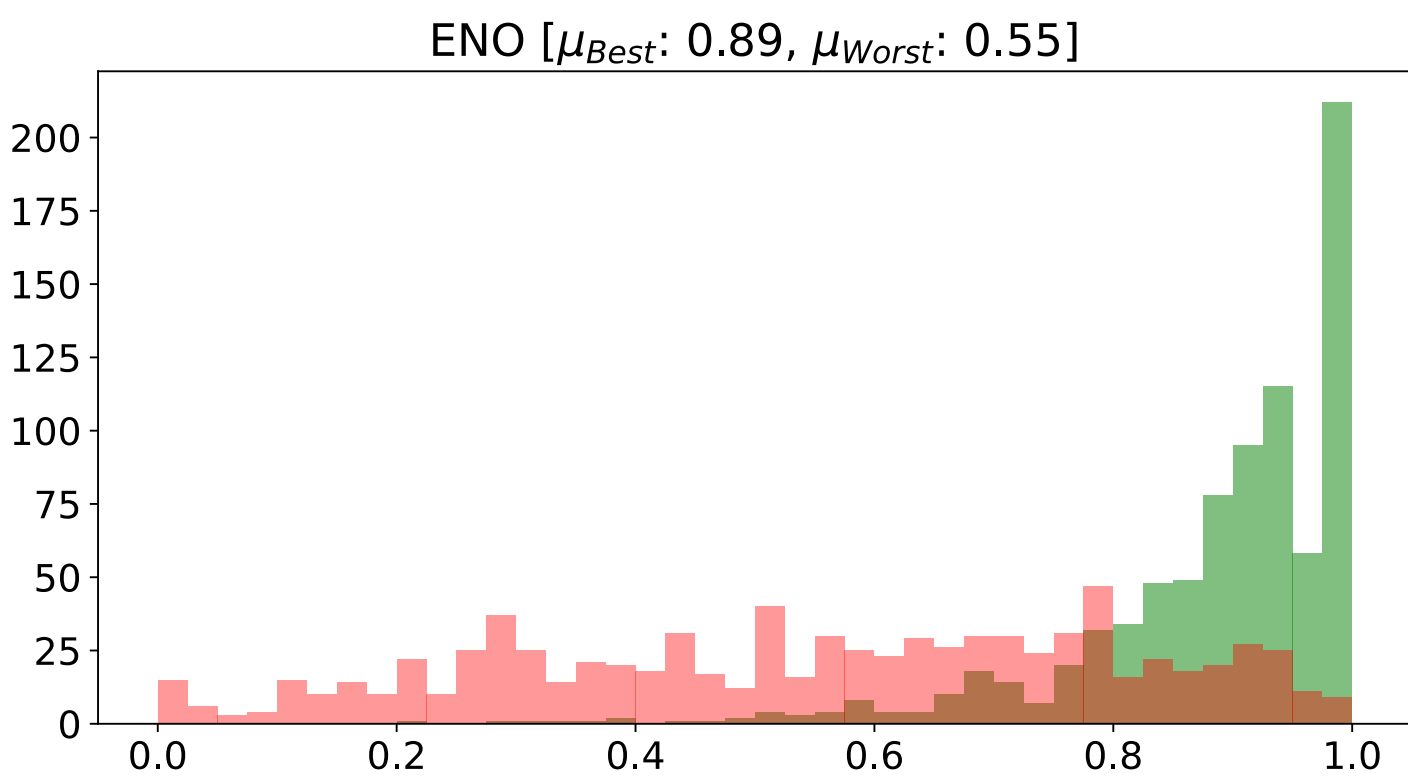
Results: Aesthetics



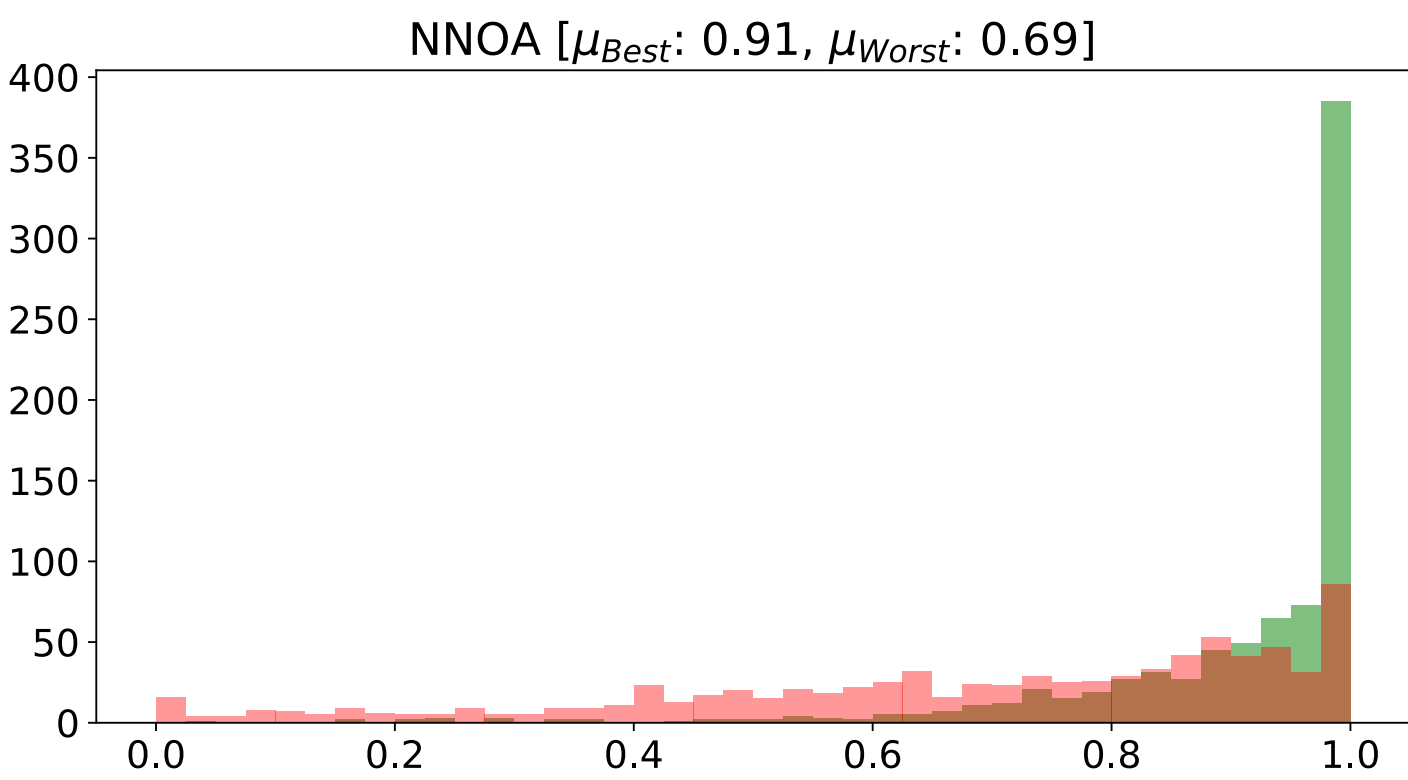
Node-Node Overlap



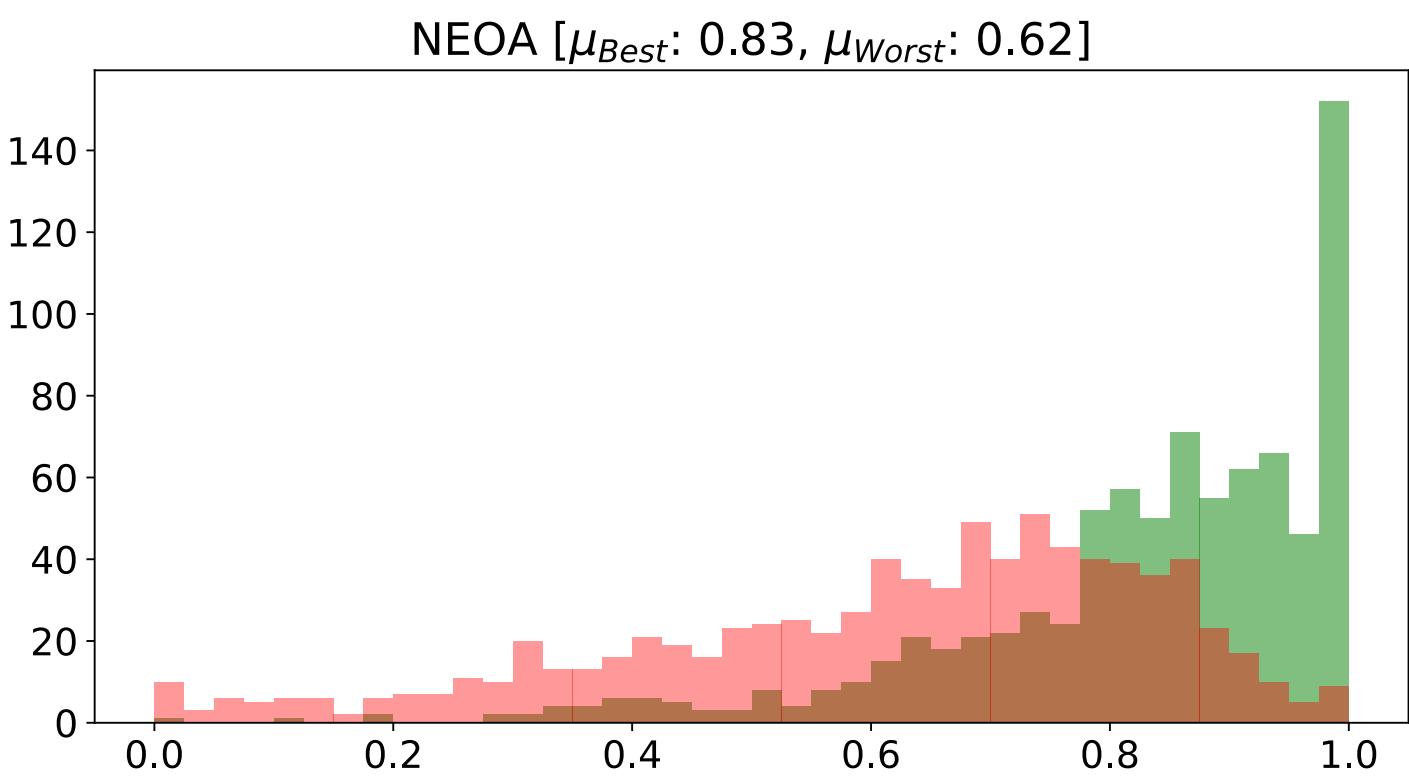
Node-Edge Overlap



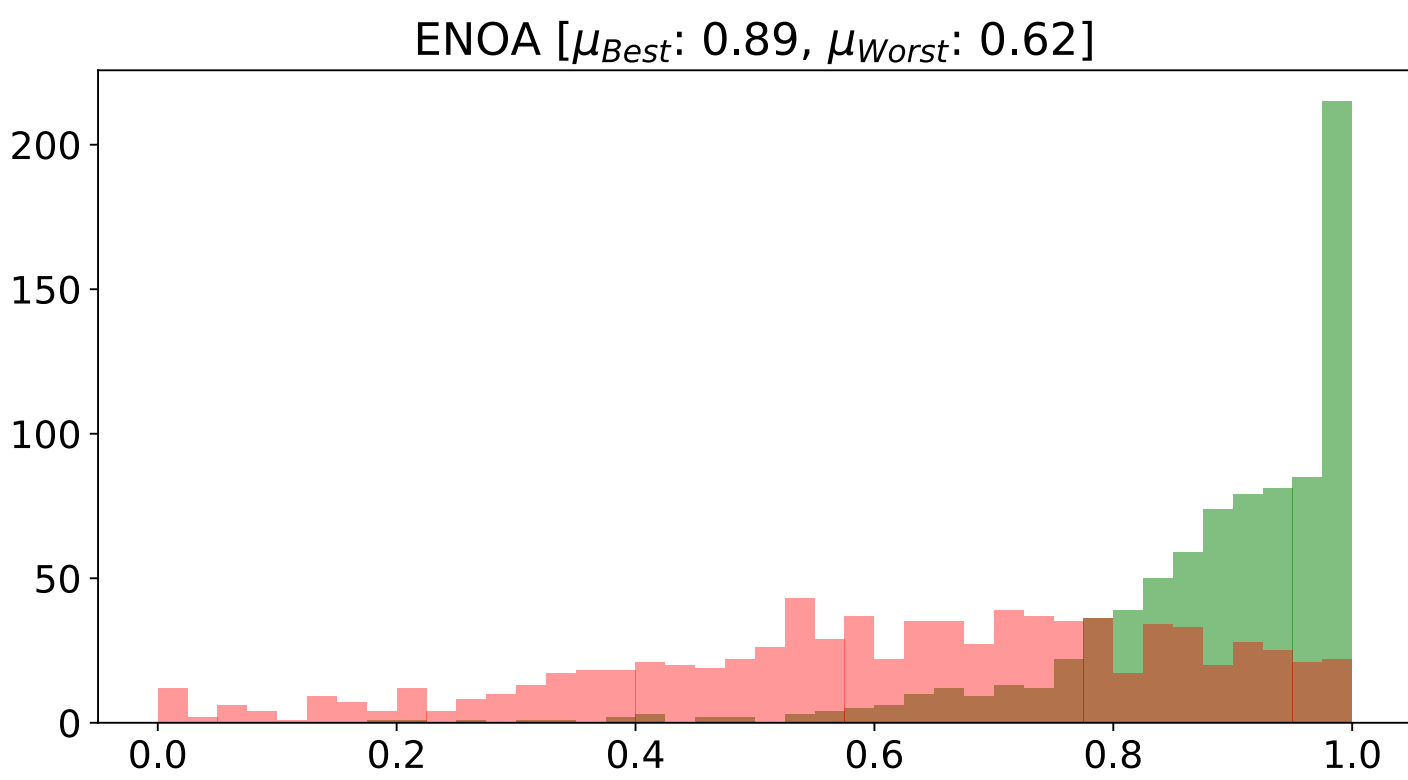
Edge-Node Overlap



Node-Node Overlap Area

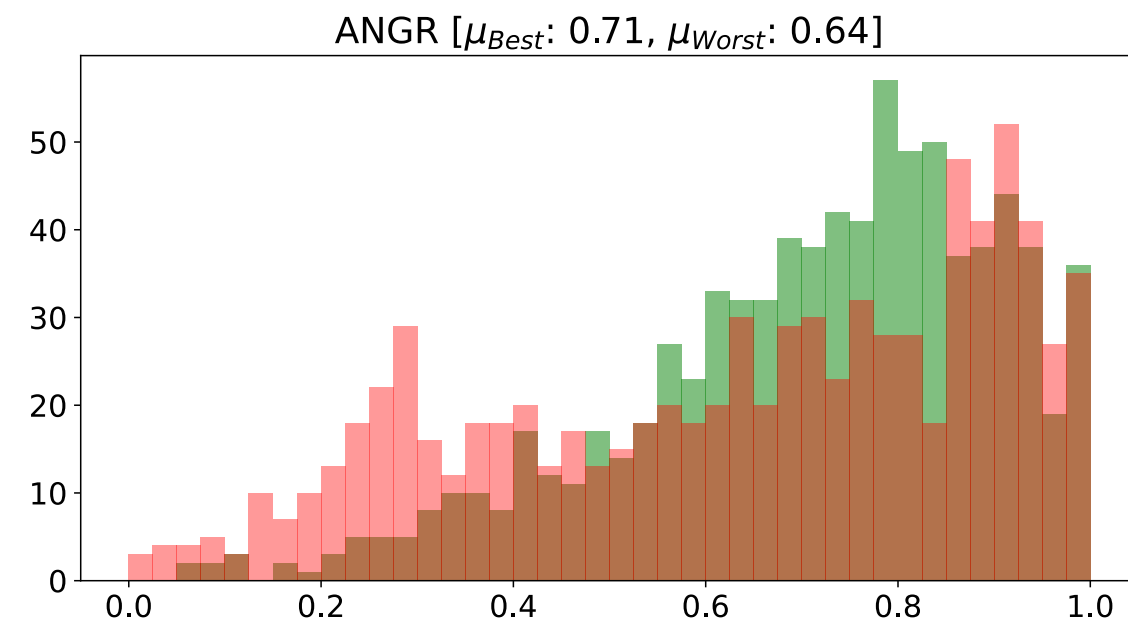


Node-Edge Overlap Area

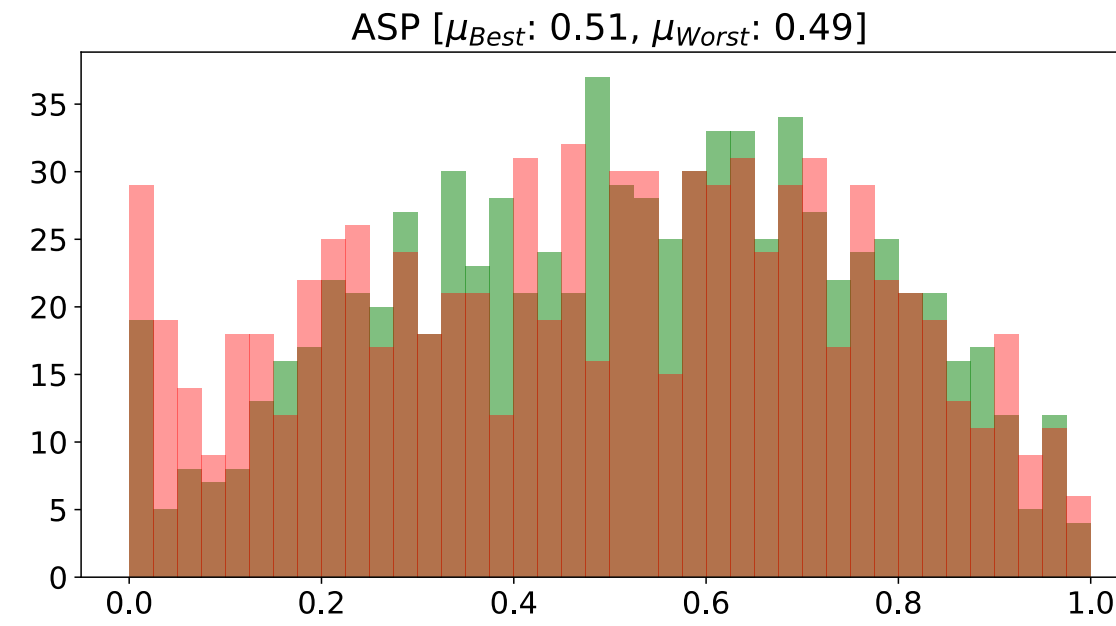


Edge-Node Overlap Area

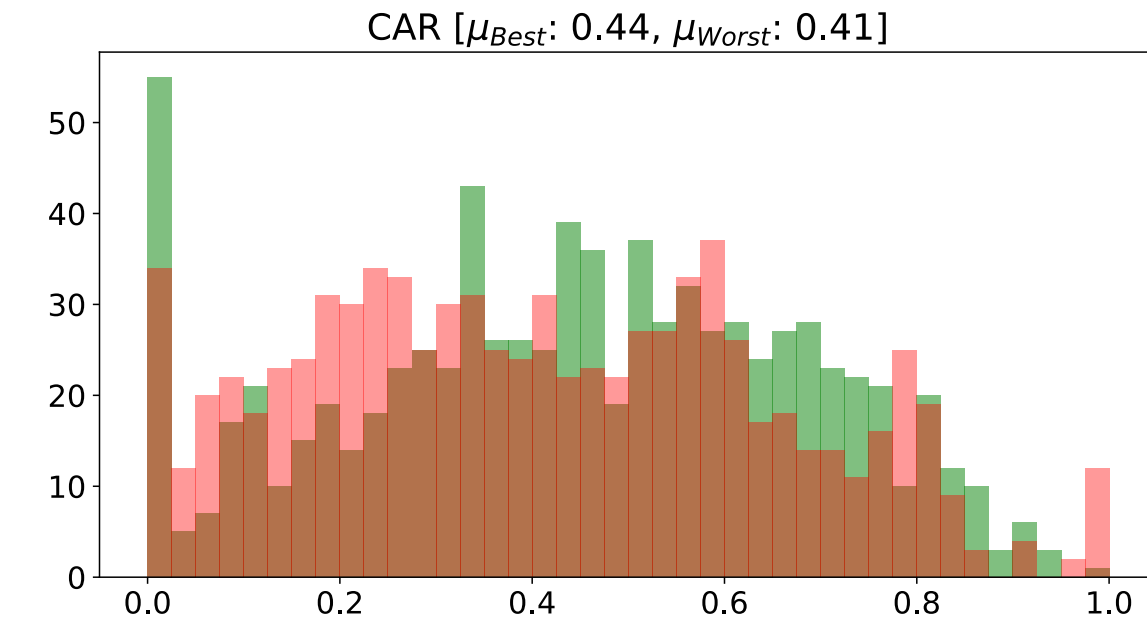
Results: Aesthetics



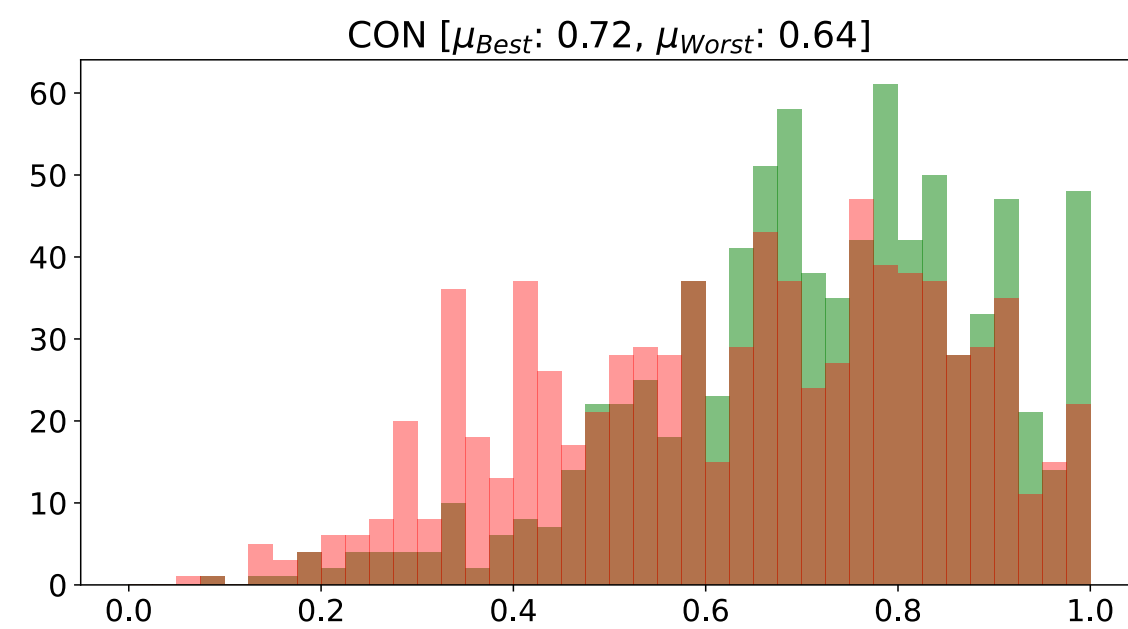
Angular Resolution



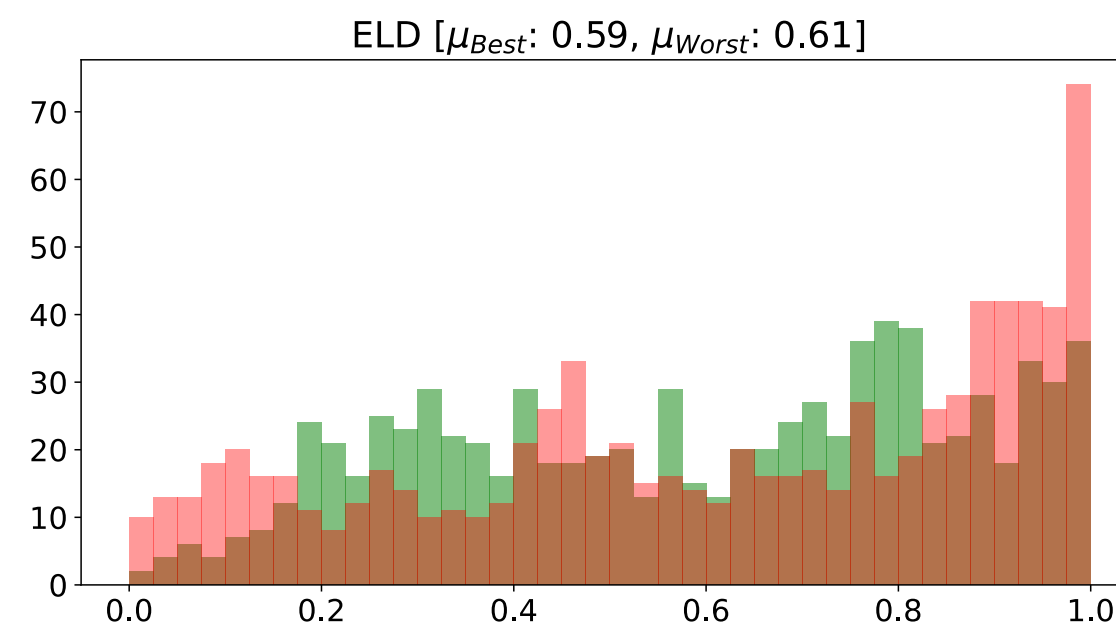
Aspect Ratio



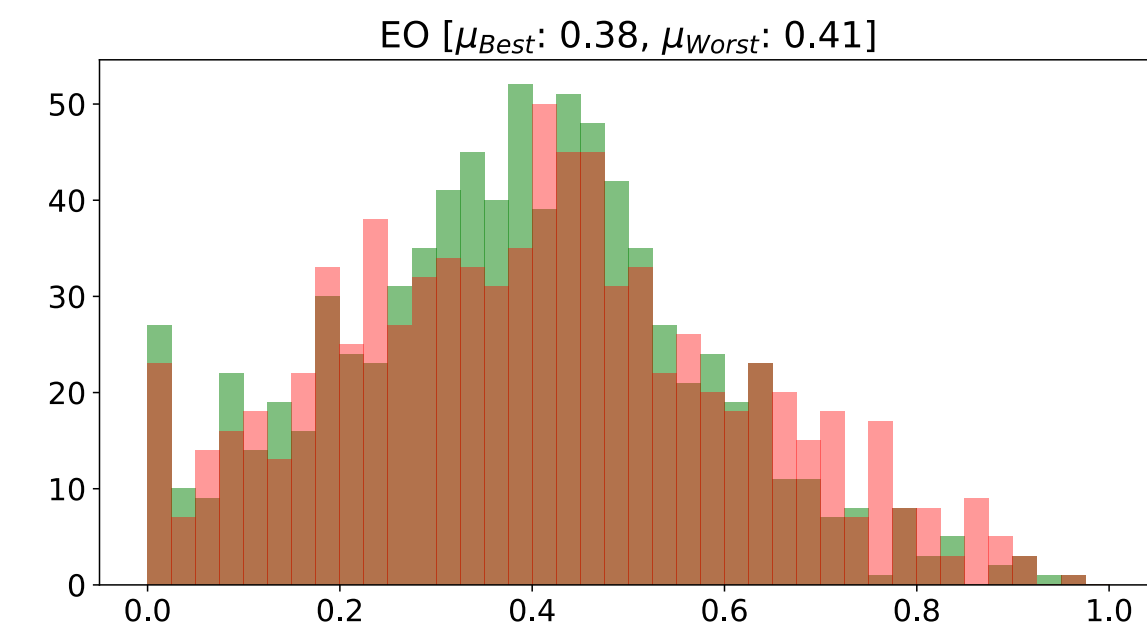
Crossing Angular Resolution



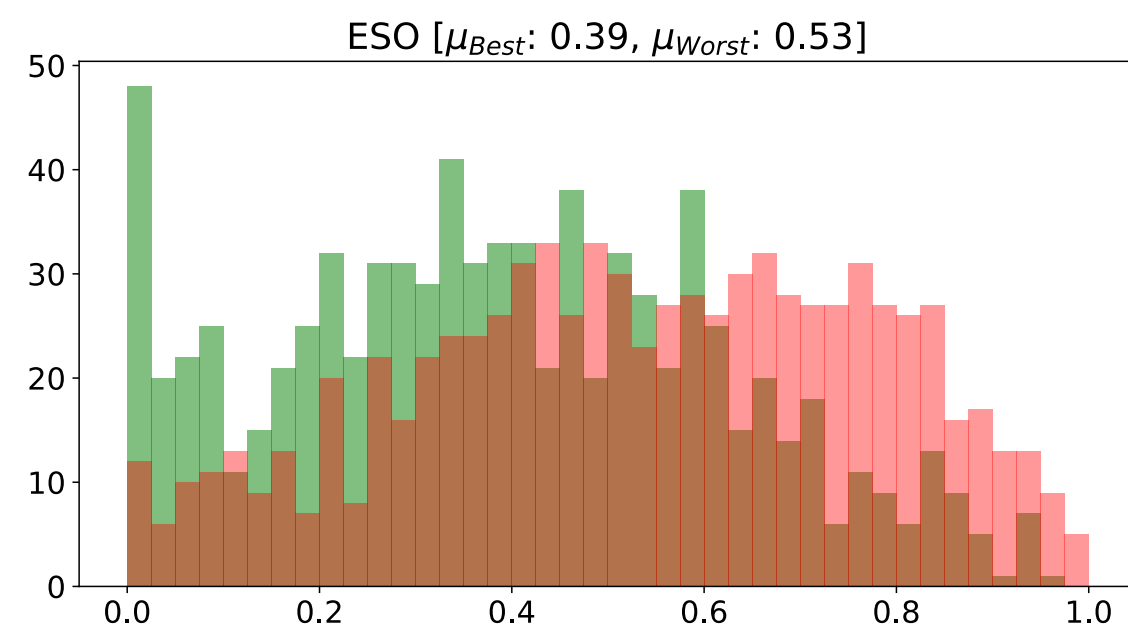
Node Concentration



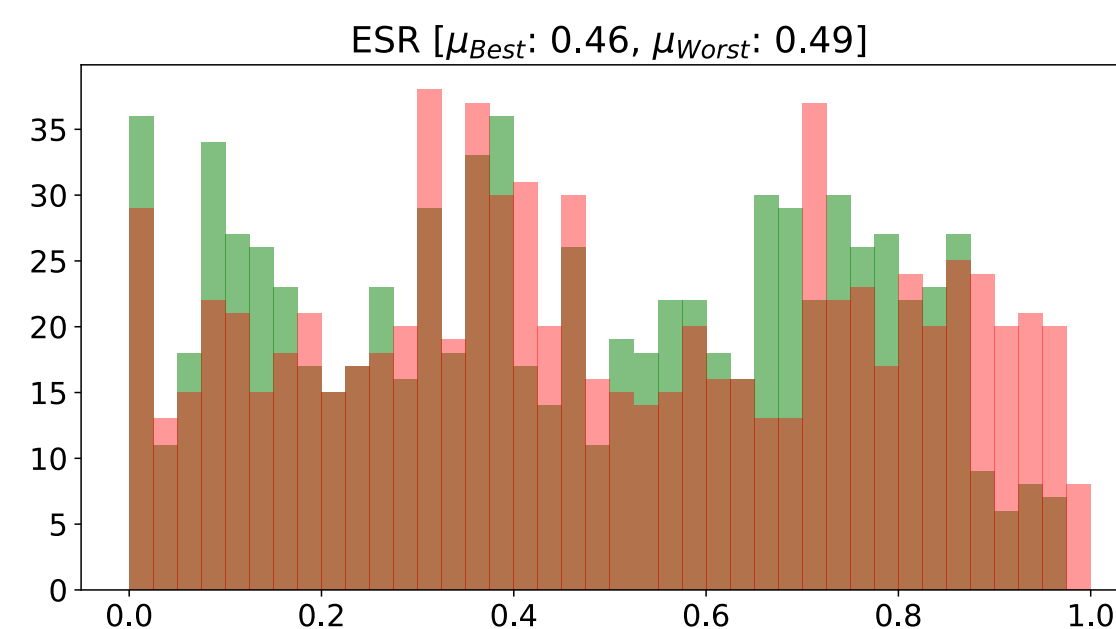
Edge Length Deviation



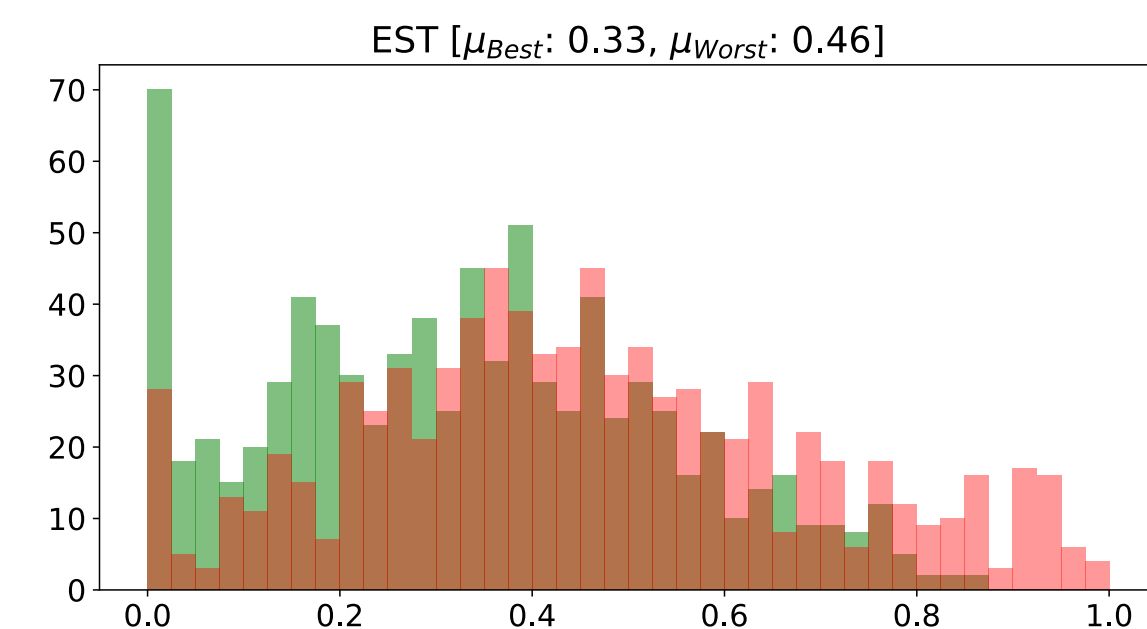
Edge Orthogonality



Edge Rotational Symmetry



Edge Reflective Symmetry



Edge Translational Symmetry

Results: Combinations

- ▶ Aim: find **combination** of aesthetics explaining **user behaviour** best
- ▶ Focus on **linear** combinations (interpretability and reduced overfitting risk)
- ▶ Two approaches:
 - ▶ Logistic Regression (**LR**): **classification**-based
 - ▶ Sequential Quadratic Programming (**SQP**): optimises **separation**
- ▶ Core **influential** measures: CR, ST, GR, ISO, overlap measures
- ▶ SQP favours **spread-out** measures (ST, GR), LR favours **separations** (ISO, overlaps)

Results: Combinations

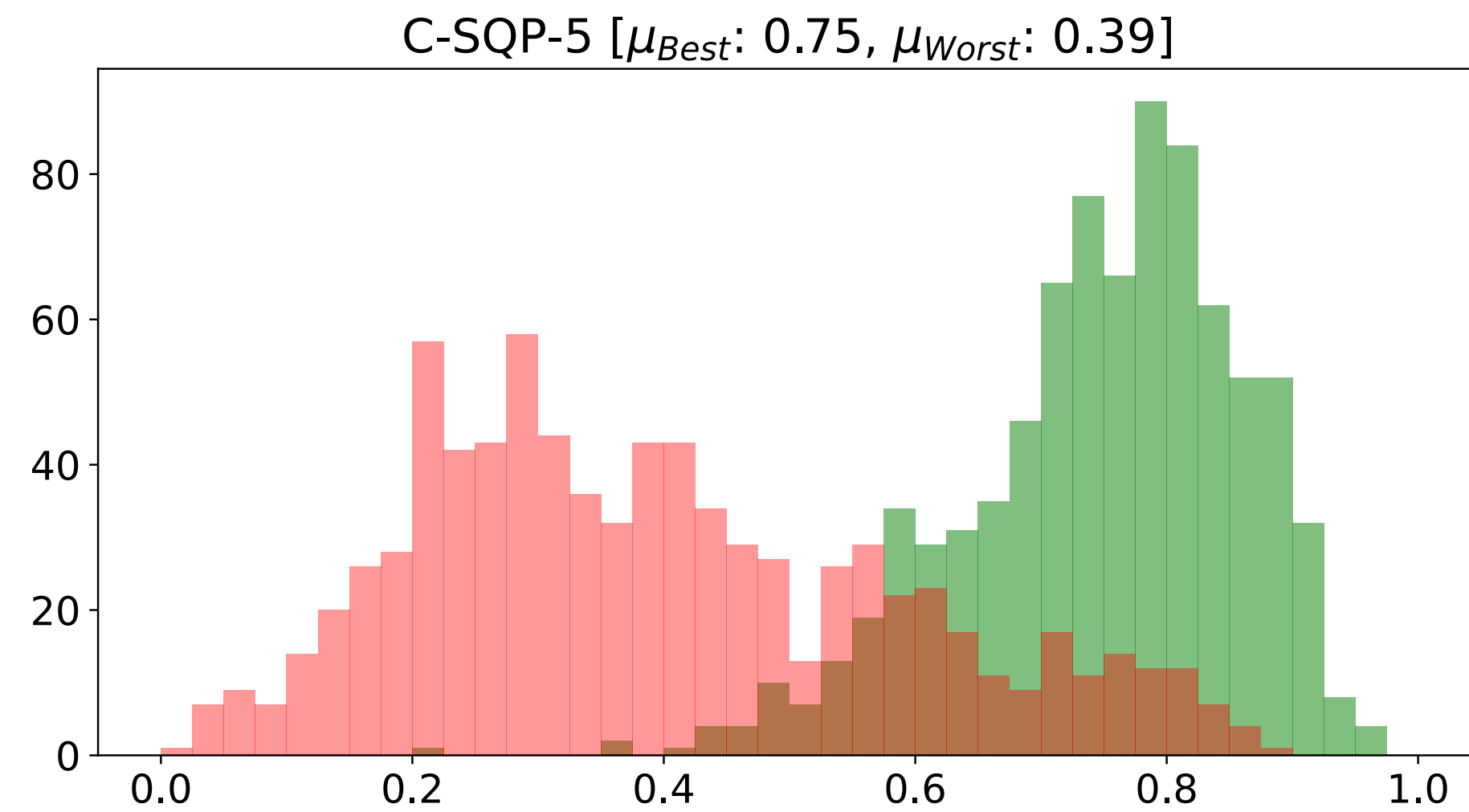
A	Data	Method	NNO	NNOA	ENO	ENOA	NEO	NEOA	CR	ST	CAR	AR	ASP	CON	NO	GR	ANGR	EO	ELD	ESR	ESO	EST	ISO
5	All	LR			.32		.19		.22											.03			.24
		SQP			.19				.20	.23						.22							.16
	L-S	LR		.21		.29			.11			.24											.15
		SQP	.18		.22	.18				.22						.20							
	L-L	LR				.13	.36						.10					.05					.35
		SQP					.17		.24	.17						.22							.20
	L-E	LR			.23	.13				.32						.24		.07					
		SQP			.17		.16		.22	.25						.21							
	S-S	LR			.32		.29		.22			.16							.01				
		SQP	.20		.21		.20			.21						.19							
	S-M	LR	.21		.37						.13					.02							.28
		SQP			.23	.19	.20			.20						.18							
	S-L	LR				.26			.34		.06		.03			.31							
		SQP			.16				.21	.25						.23							.15
	S-XL	LR		.12		.20			.29	.19					.20								
		SQP			.15				.23	.22						.23							.16

Overlap Measures

CR / ST / GR (correlated)

ISO

Results: Combinations



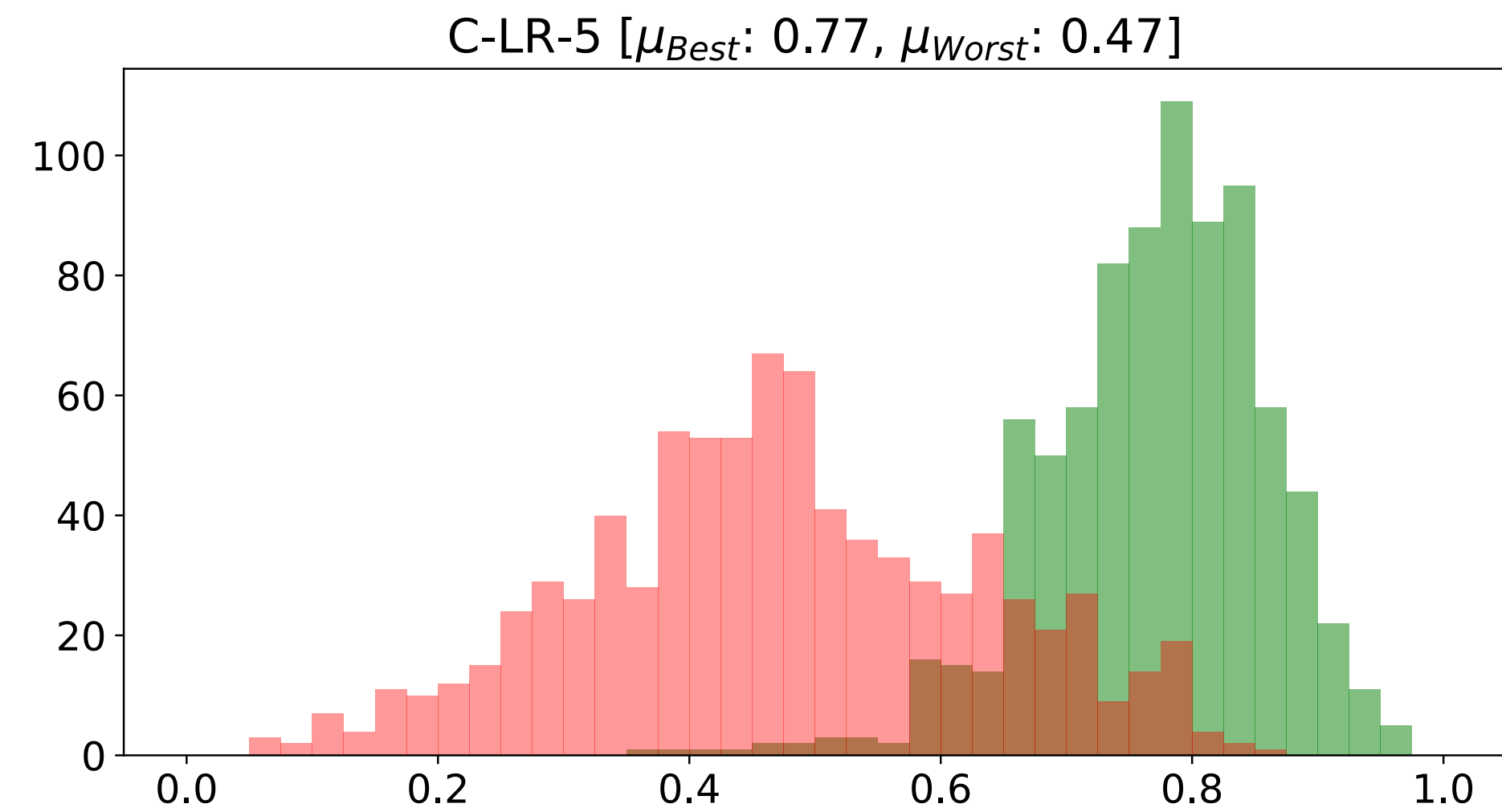
ST (0.23)

GR (0.22)

CR (0.20)

ENO (0.19)

ISO (0.16)



ENO (0.32)

ISO (0.24)

CR (0.22)

NEO (0.19)

ESR (0.03)

Limitations & Future Work

▸ Limitations:

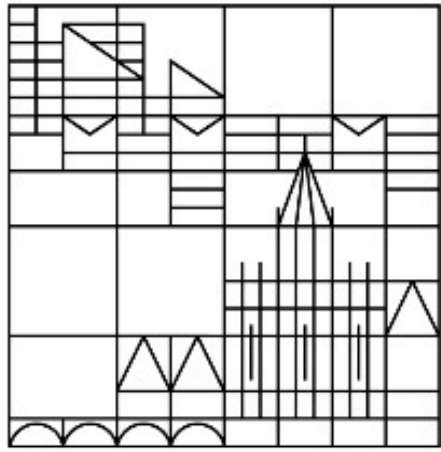
- Limited graph **sample**
- Participant pool mainly **novices**, results may differ for experts
- Focused on **preferences** rather than **task-solving** performance
- Treated viewpoints as **static**, not considering navigation or transitions

▸ Future Work:

- Viewpoint stability (**landscape**)
- Relation to **task-solving** performance
- Expand participant diversity, including **expert** users
- Develop and validate new **3D-specific aesthetics** (e.g., building on ISO measure)

Conclusion

- ▶ **Study** with 23 participants **on viewpoint preferences** across diverse **3D graphs**
- ▶ User choices balance multiple cues: **classical 2D measures** (Stress, Crossings, Gabriel Ratio) and **3D-specific** ones (Edge-Node Overlap, Isometric Deviation)
- ▶ **Global structure** often prioritised over avoiding local overlaps or crossings
- ▶ A **small set** of measures explains preferences well
- ▶ **Open dataset** enables further research on viewpoint evaluation and optimisation



Data Analysis
and Visualization



Show Me Your Best Side:

Characteristics of User-Preferred Perspectives for 3D Graph Drawings

Thank You!
Questions?